

Diffusion in networks

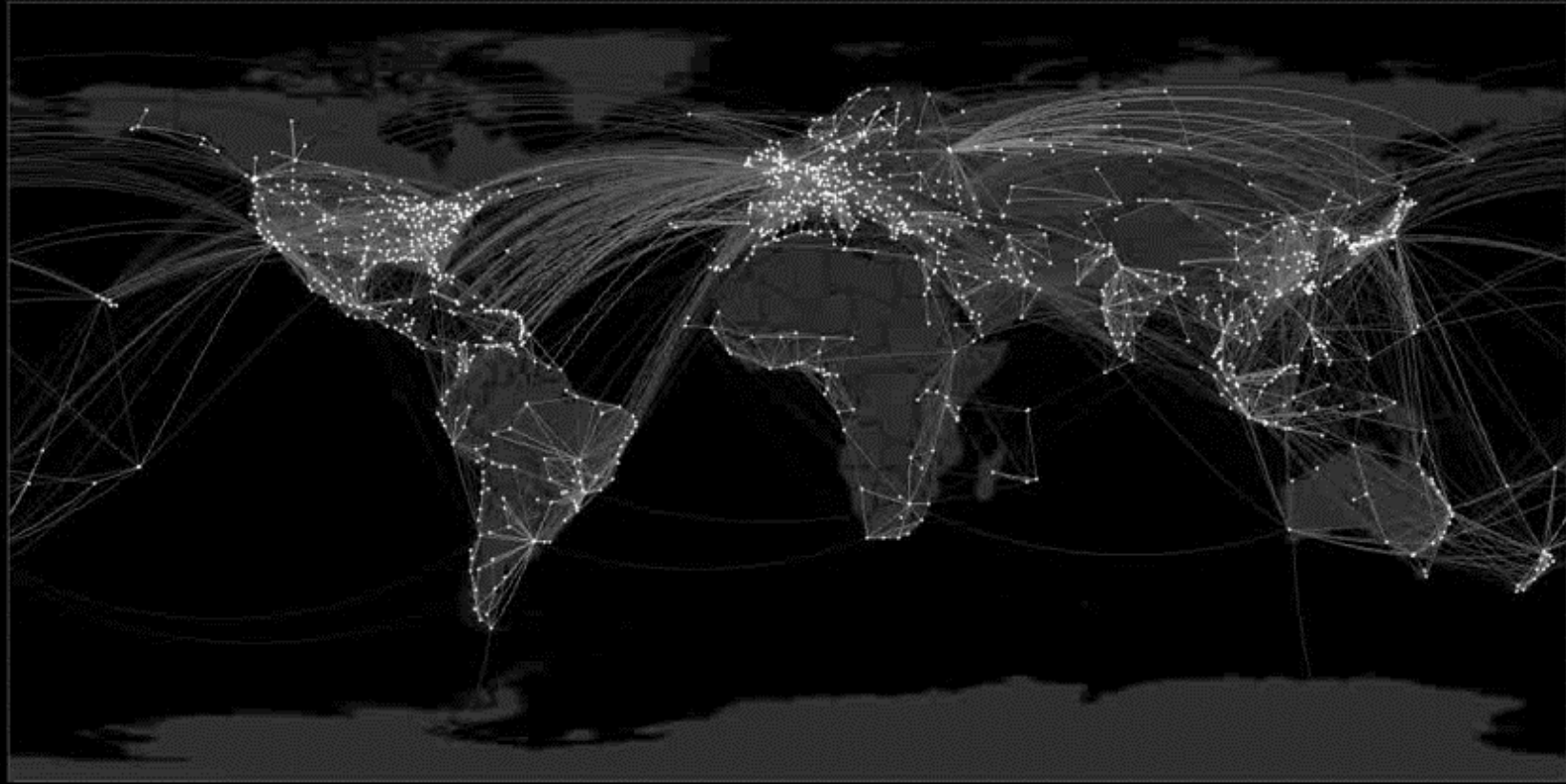
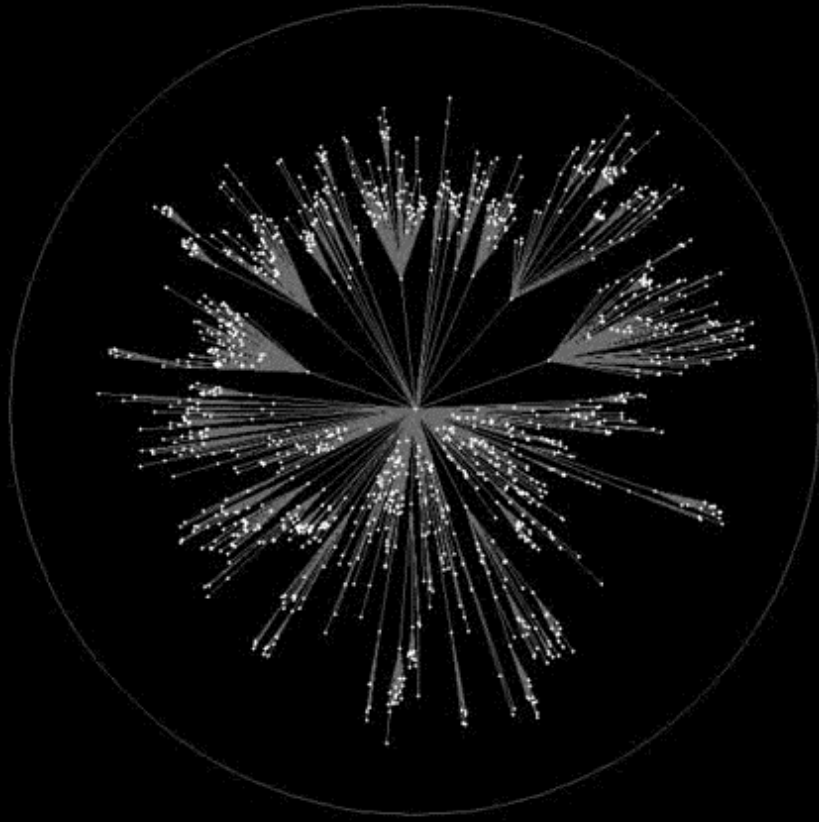
Lengyel Balázs

ANETI Lab, ELTE KRTK KTI and CIAS

IDAIS CUB

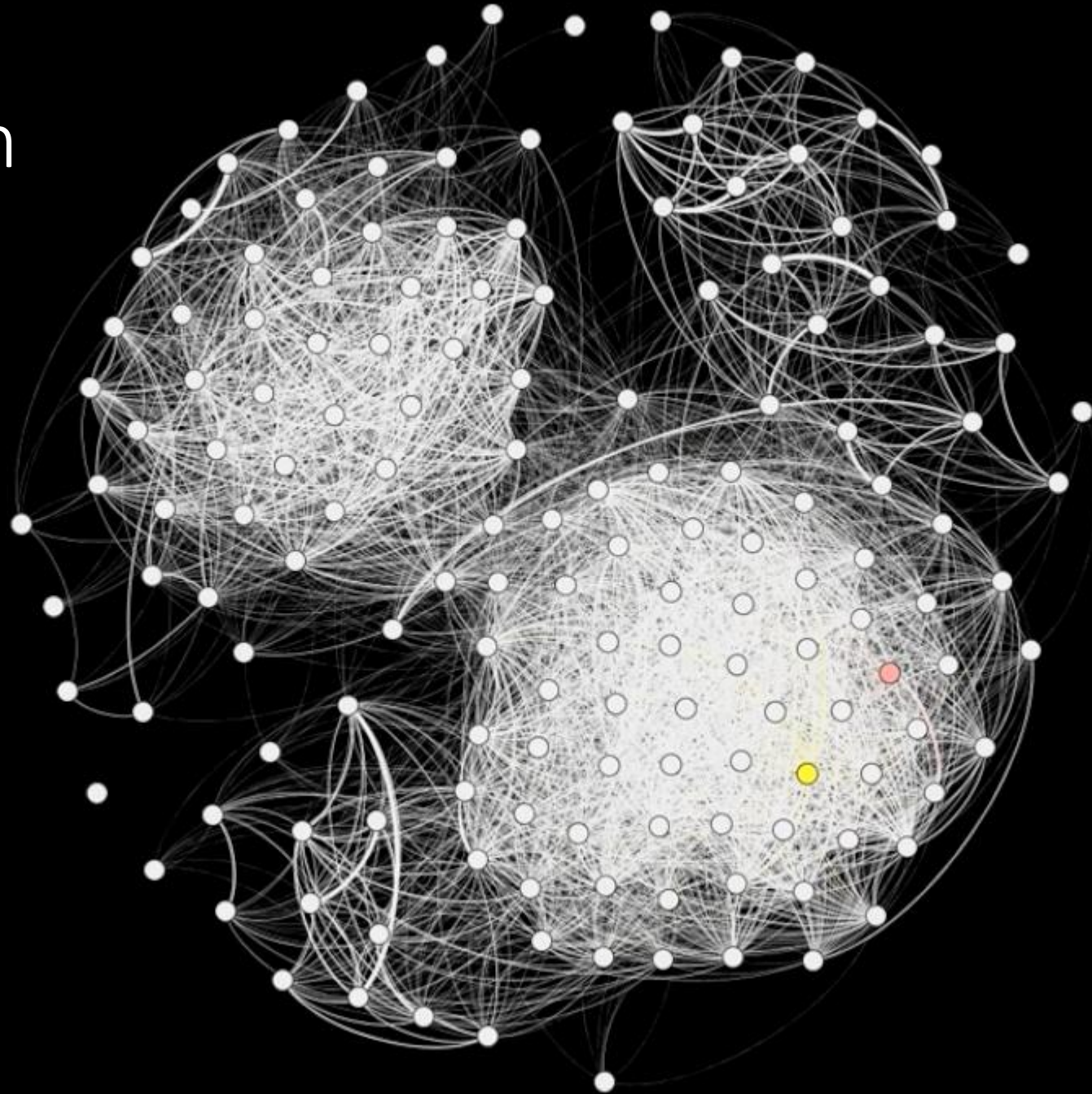
2025. Nov 25.

Virus diffusion on the global flight network



Brockman D., Helbing D (2013) The hidden geometry in complex, network-driven contagion phenomena. Science 342 (6164) pp. 1337-1342.

Diffusion in networks



Today

I. Diffusion in networks

1. Diffusion models: Bass modell {innovation} and SI models {virus} (SIS, SIR)
2. The impact of network structure on diffusion process
3. Simple versus Complex diffusion: spreading of viruses versus innovation

II. Shock propagation and adaptation over crisis

1. Propagation in economic networks
2. Network dynamics

III. The spatial diffusion of innovation in networks / EC material

I. 1. Diffusion models

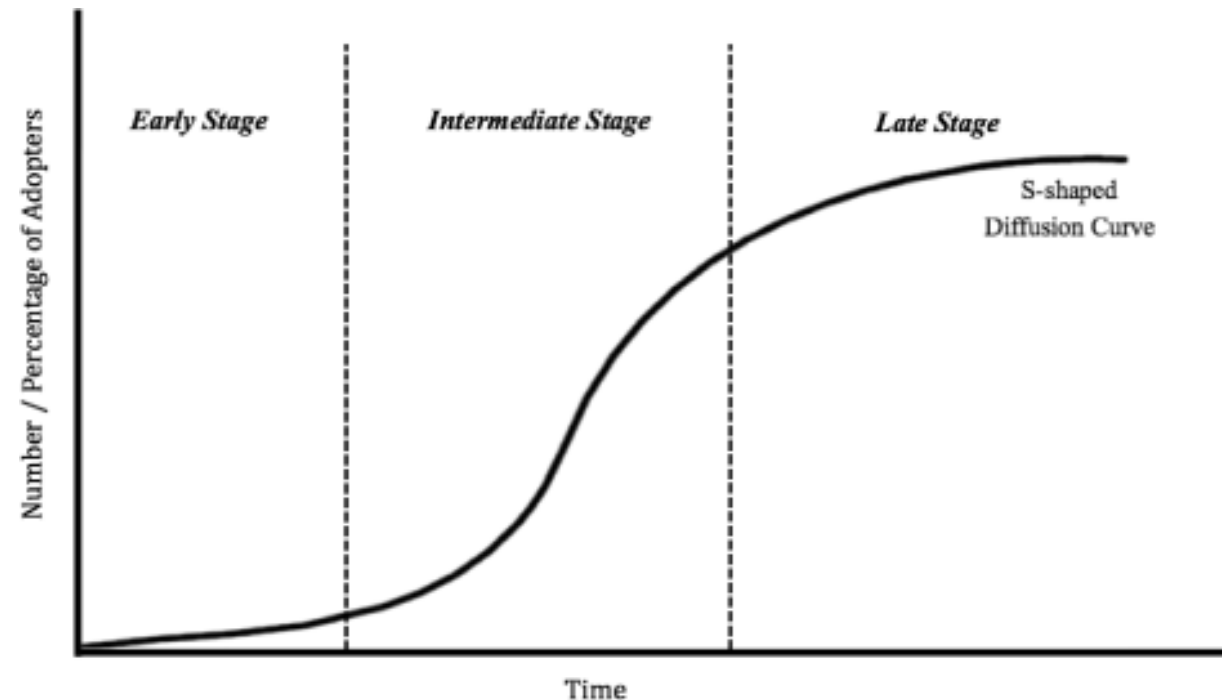
The S curve of diffusion

Coleman, Katz, Menzel (1966): how fast do doctors prescribe doctors new medicins?

- diffusion: prescriptions
- network: survey information about advice links between doctors
- High degree doctors started to prescribe earlier.

Griliches hybrid corn example: how big fraction of farmers use it?

- Initially slow diffusion accelerates and slows down again.



Bass model

Two states / activity: 0 or 1 – yes or no

- $F(t)$ – fraction of the population who started the activity until time t .
- p is the rate of spontaneous adoption
- q is the rate of imitation

$$\frac{dF(t)}{dt} = (p + qF(t)) (1 - F(t))$$

- Second term: fraction of population who have not adopted until time t .
- They can adopt spontaneously or follow the adopters.
- Solution to differential equation is:

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}}$$

Bass model

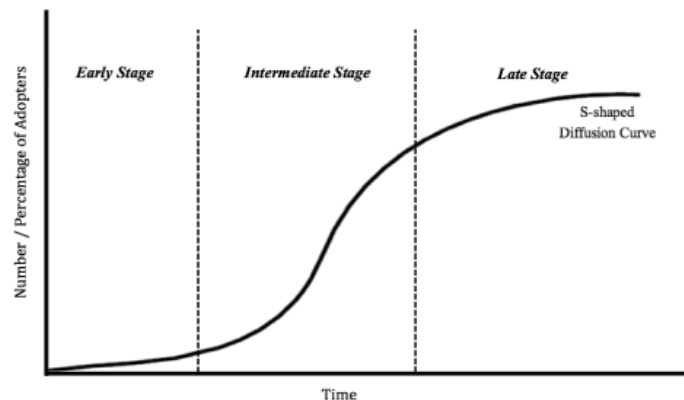
If $F(t)$ is close to 1, diffusion slows down

- Second term goes to 0, growth goes to 0

$$\frac{dF(t)}{dt} = (p + qF(t))(1 - F(t))$$

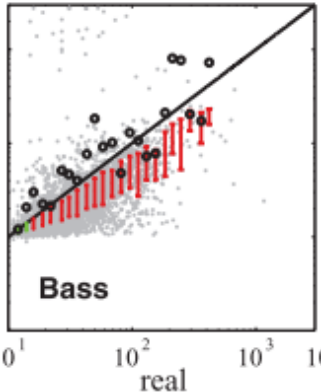
In the beginning, only p matters, but later q becomes more important

- In the beginning $F(t)$ is small, thus $qF(t)$ goes to 0
- There is no one to adopt but anyone can adapt spontaneously
- The slope of spread is p in the beginning, but then q speeds this up.

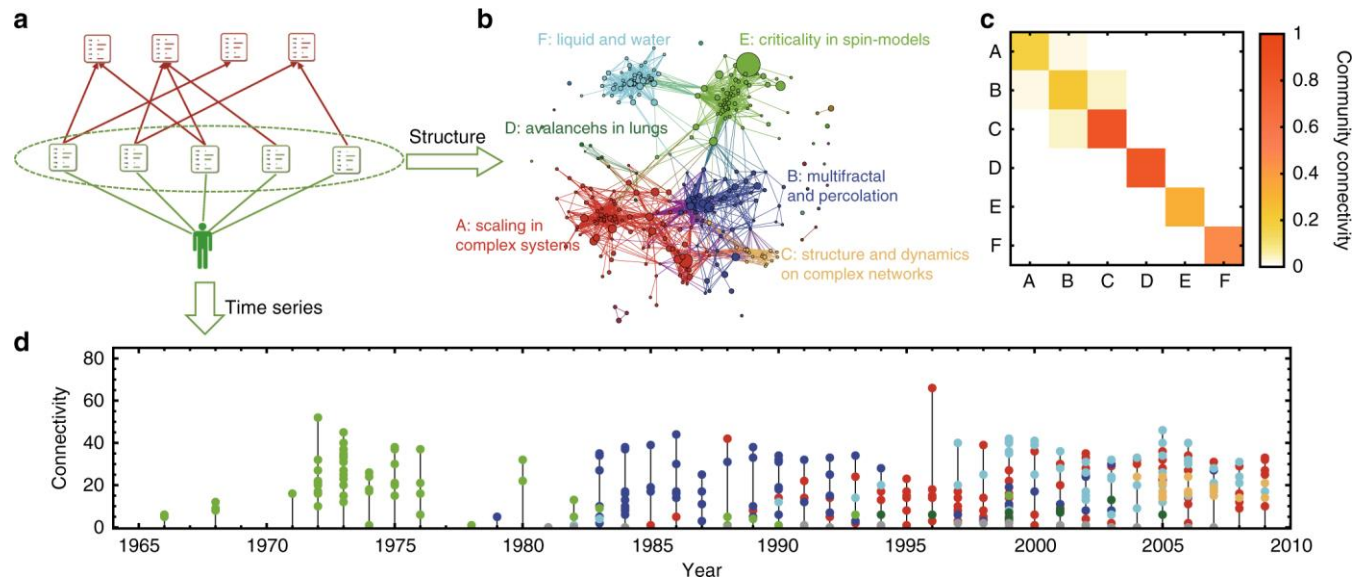


Bass model research to predict impact (the LI70 hypothesis)

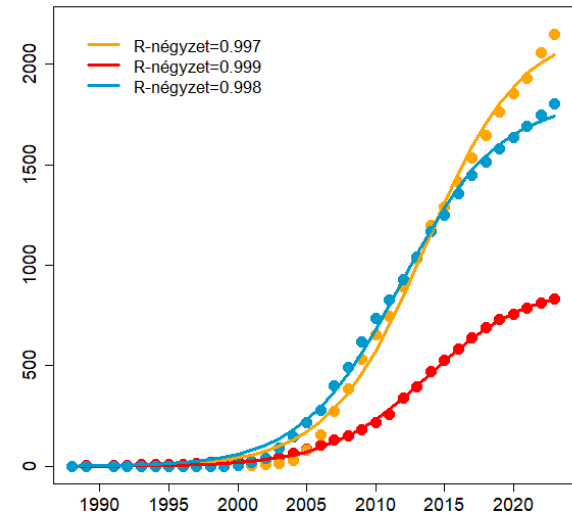
p and q can be estimated early and then predictions can be made



Wang et al. 2013 *Science*



Zeng et al. 2019 *Nature Communications*

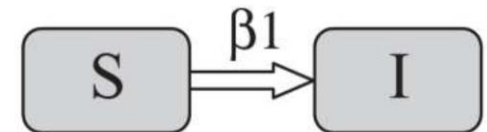


LI70 hypothesis: „Changes in research topics along the academic career make the prediction of success more difficult.”

SI model

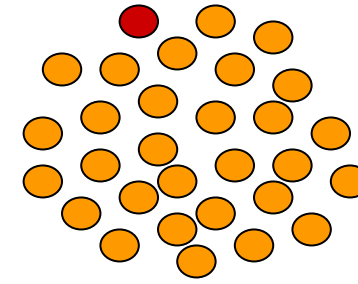
- N : size of population
- S : susceptible population. Those who can be potentially infected.
- I : infected population. Those who can infect others.
- $S(t)$: susceptibles at time t .
- $I(t)$: infected at time t .
- β : probability of transmission

$$N = S(t) + I(t)$$



SI model

- The likelihood that the disease will be transmitted from an infected to a healthy individual in a unit time: β
- In every period t Infected individual will infect βS individual.
- In $t+1$, $\beta I S$ individual will be infected.
- Each individual has $\langle k \rangle$ contacts with randomly chosen other individuals per unit time.
- Transmission rate: $\beta \langle k \rangle$



If there are I infected individuals and S susceptible individuals, the average rate of new infection is $\beta S I / N$

$$\frac{dI(t)}{dt} = \beta \langle k \rangle \frac{S(t)I(t)}{N}$$

SI model

$$s = S/N, \quad i = I/N$$

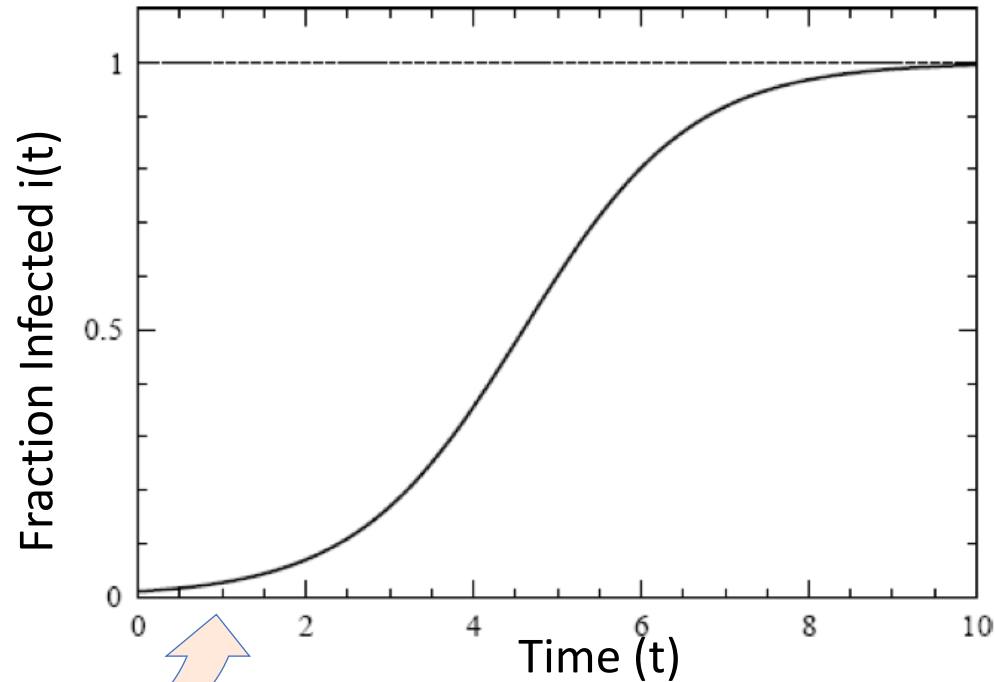
$$\frac{di}{dt} = \beta s i = \beta i(1 - i)$$

If $i(t)$ is small,

$$\frac{di}{dt} \approx \beta i$$

$$i \approx i_0 \exp(\beta t)$$

**exponential
outbreak**



As $i(t) \rightarrow 1$.

$$\frac{di}{dt} \rightarrow 0$$

saturation

SI model: the fraction infected increases until everyone is infected.

SIS model

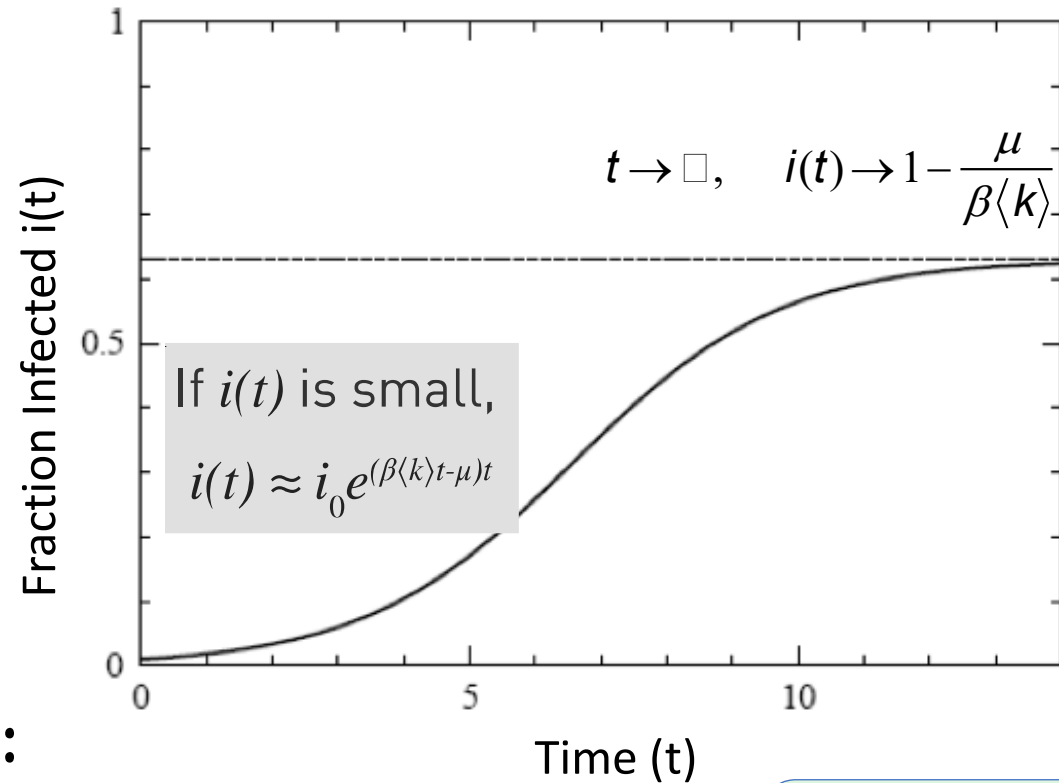
Recovered individuals become Susceptibles again.

$$\frac{di}{dt} = \beta \langle k \rangle \underset{\text{I}}{i} \underset{\text{S}}{(1-i)} - \underset{\text{I} \rightarrow \text{S}}{\mu} i$$

$$i = \left(1 - \frac{\mu}{\beta \langle k \rangle}\right) \frac{C e^{(\beta \langle k \rangle - \mu)t}}{1 + C e^{(\beta \langle k \rangle - \mu)t}}.$$

Endemic state ($\mu < \beta \langle k \rangle$):

Disease-free state ($\mu > \beta \langle k \rangle$):



Stationary state:

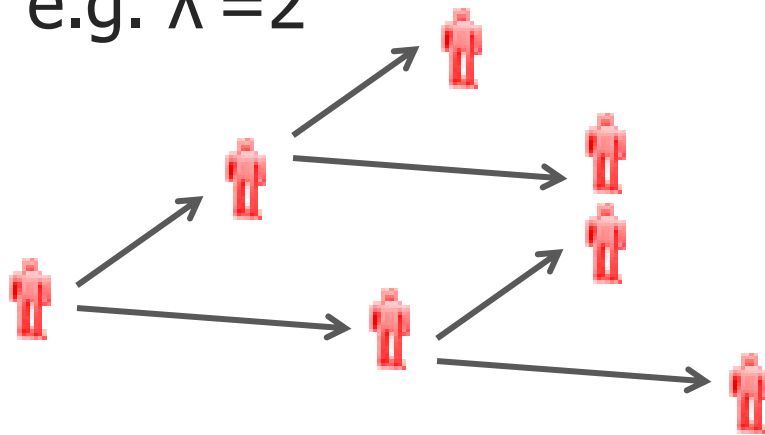
$$\frac{di}{dt} = \beta i(1-i) - \mu i = 0$$

SIS model

reproductive number R_0 (λ): average # of infectious individuals generated by one infected in a fully susceptible population.

$$R_0 = \frac{\beta \langle k \rangle}{\mu}.$$

e.g. $\lambda = 2$



Choose
transmission
scenario



mild

$\lambda = 1.5$



medium

$\lambda = 1.9$



high

$\lambda = 2.3$

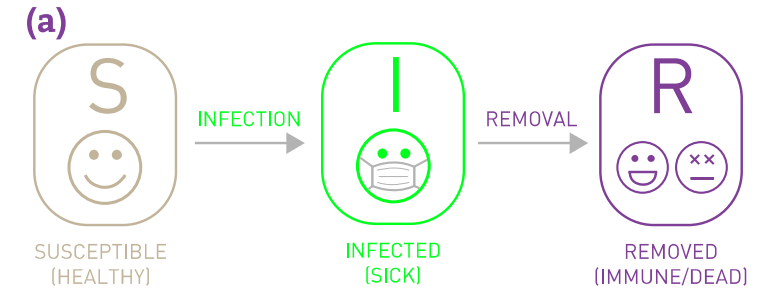


very high

$\lambda = 2.7$

SIR model

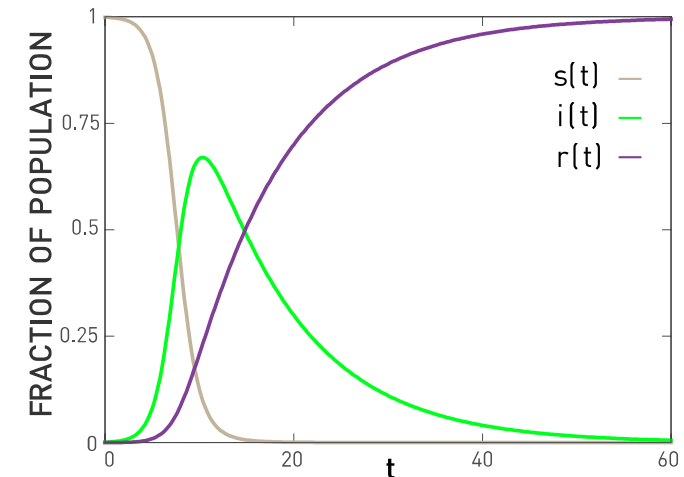
- Besides I and S, individuals have R (recovery/removal) state as well.
- Individuals cannot transform from R to I or S.
- r is the recovery probability in t .



(b)

$$\begin{aligned}\frac{ds(t)}{dt} &= -\beta \langle k \rangle i(t) [1 - r(t) - i(t)] \\ \frac{di(t)}{dt} &= -\mu i(t) + \beta \langle k \rangle i(t) [1 - r(t) - i(t)] \\ \frac{dr(t)}{dt} &= \mu i(t).\end{aligned}$$

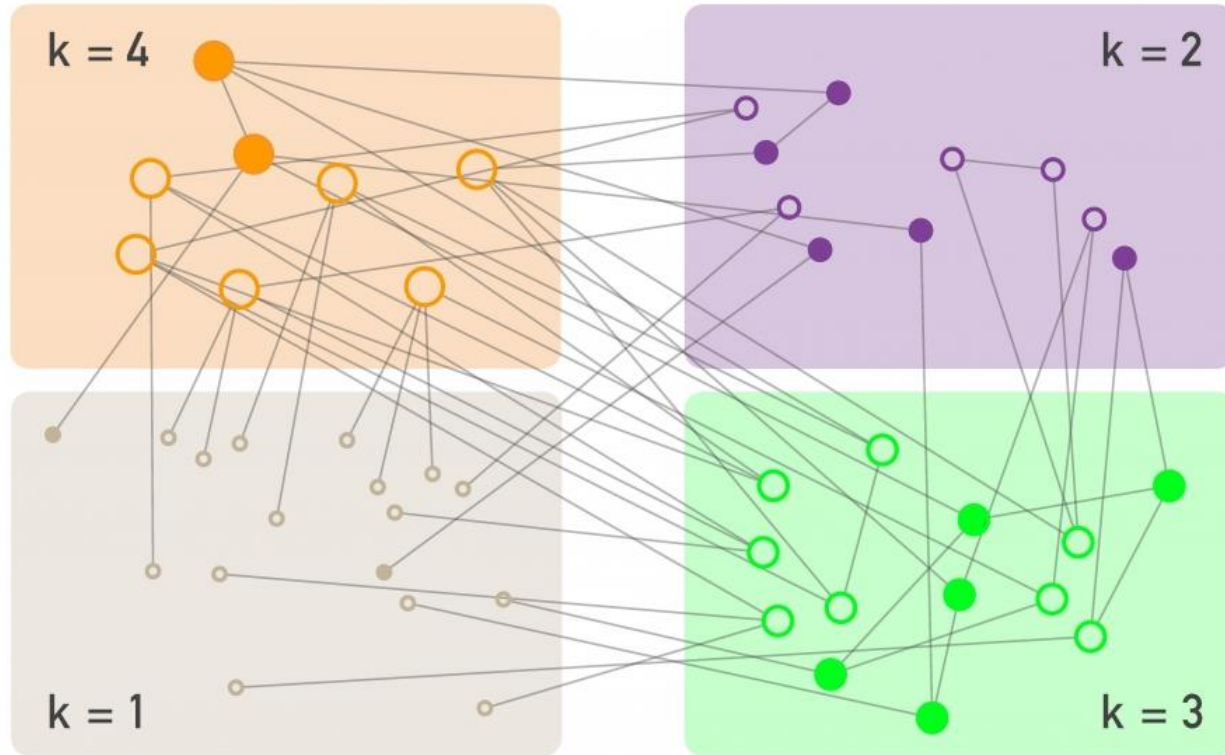
(c)



I. 2. The impact of network structure on diffusion process

SIS model on a network: degree block approximation

Split nodes by their degrees



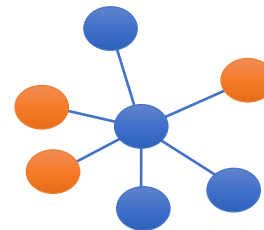
$$i_k = \frac{I_k}{N_k}, \quad i = \sum_k P(k) i_k$$

SIS model:

$$\frac{di_k(t)}{dt} = \beta(1 - i_k(t))k\Theta_k(t) - \mu i_k(t)$$

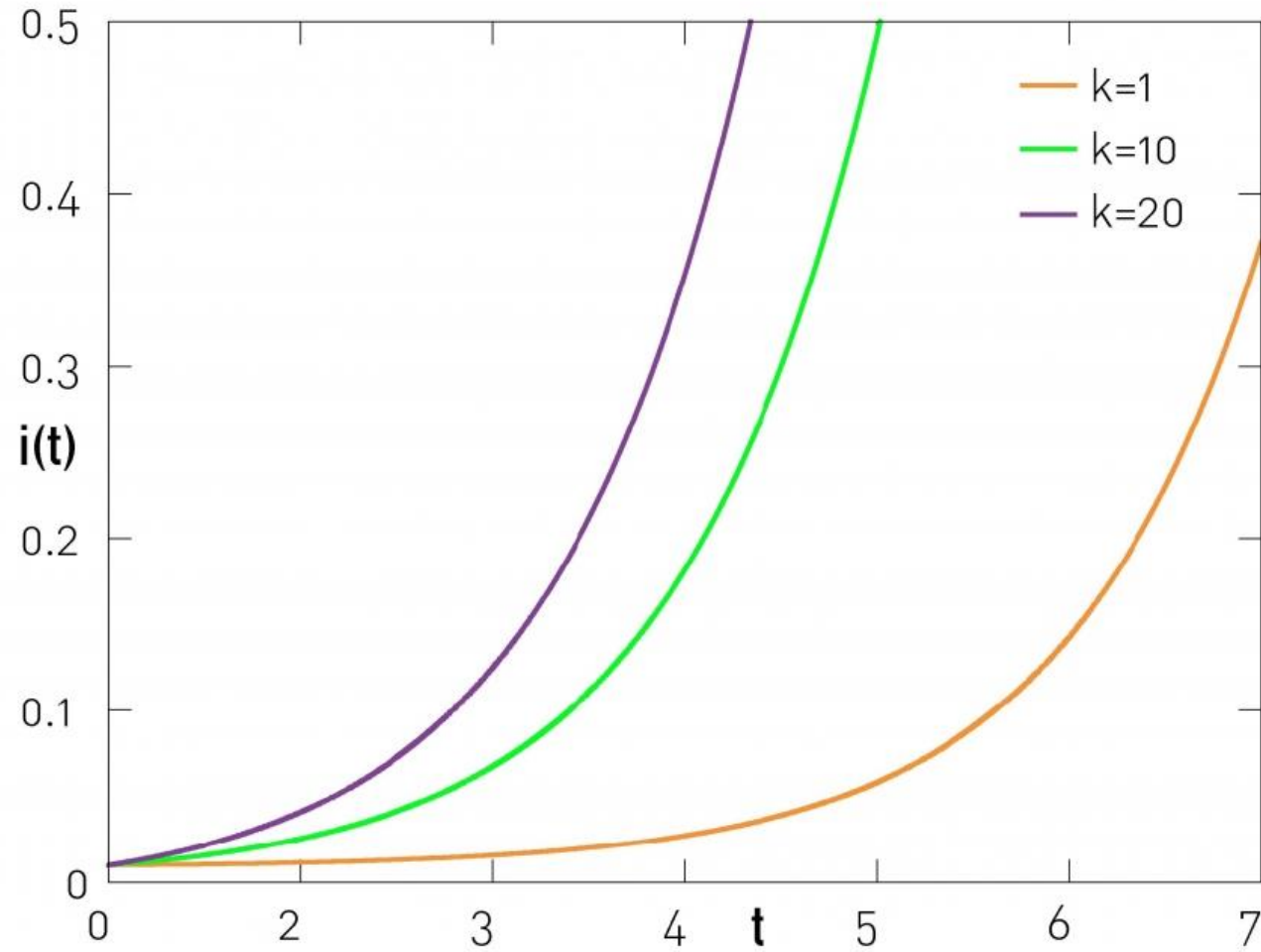
Proportional to k

Density of infected
neighbors of nodes with
degree k



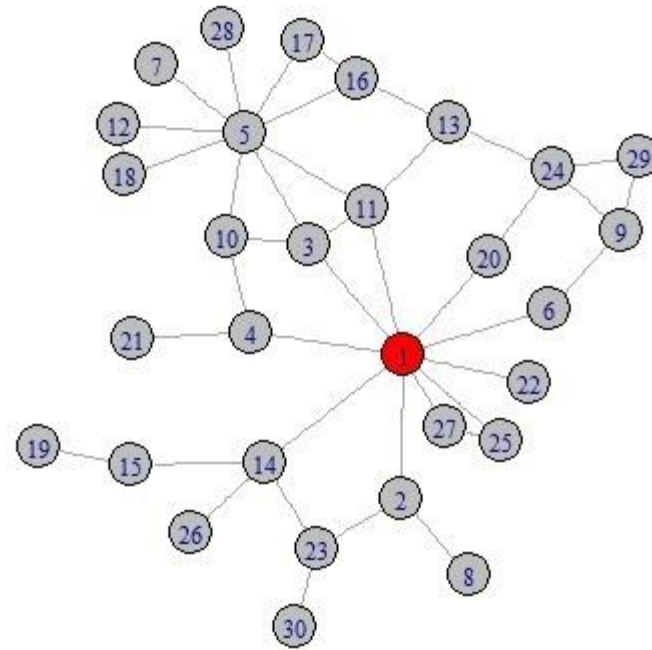
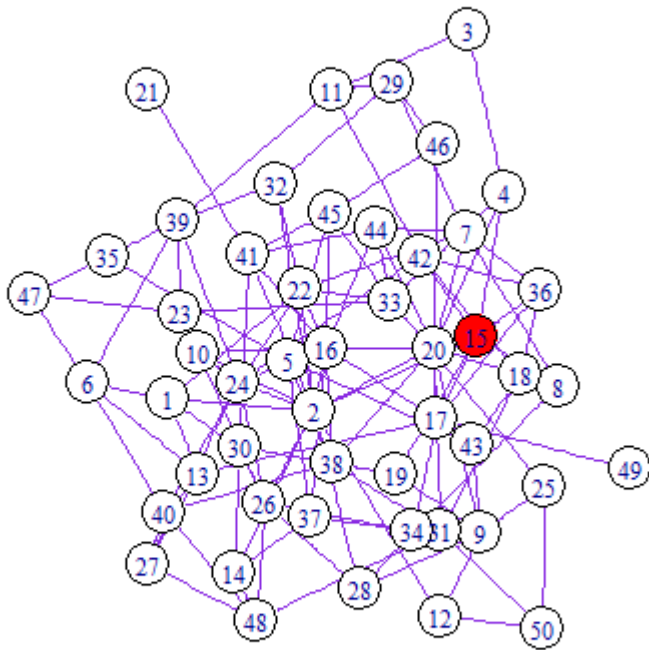
I am susceptible with k
neighbors, and $\Theta_k(t)$
of my neighbors are infected.

Degree is key in diffusion

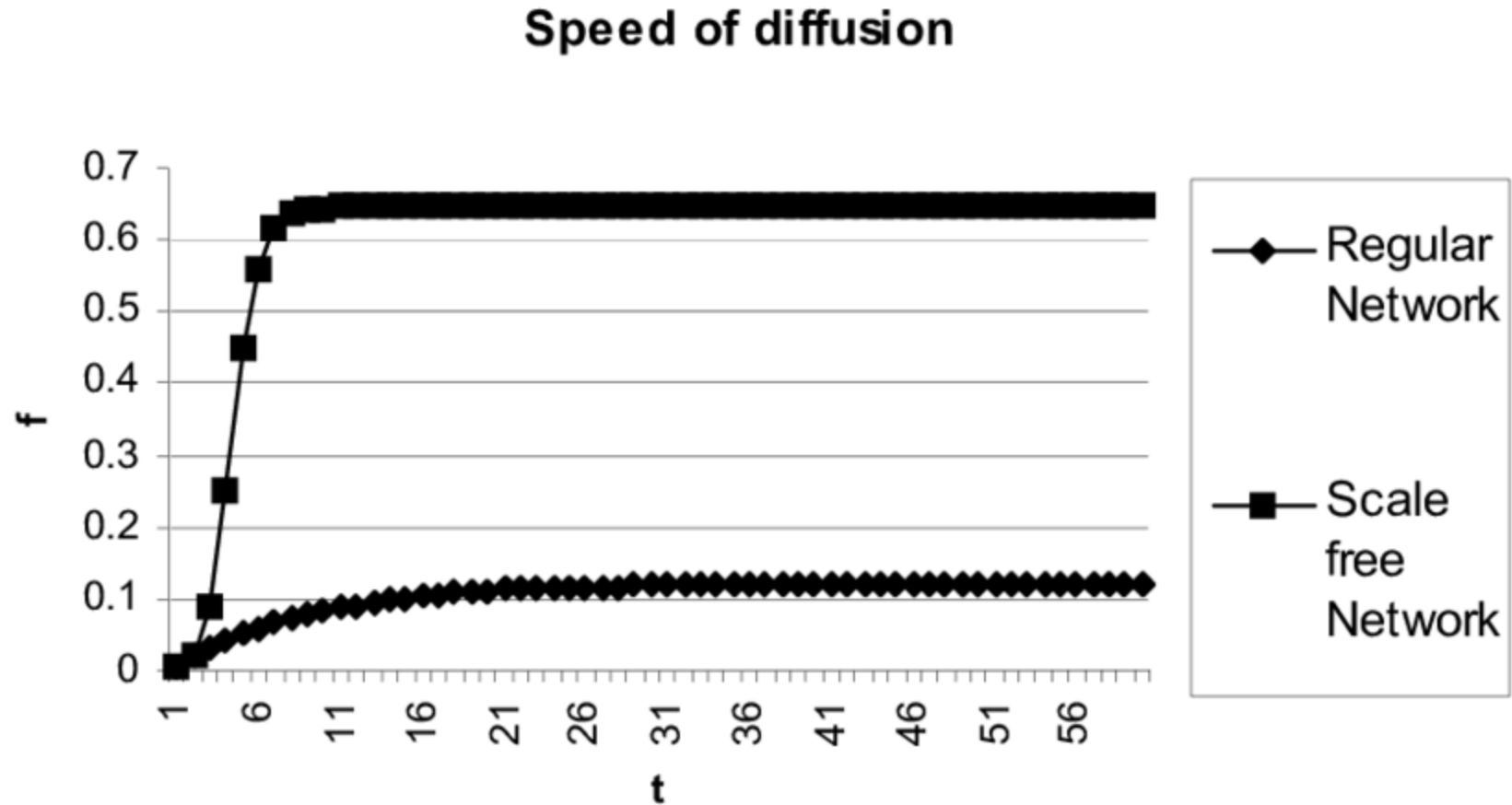


The role of hubs

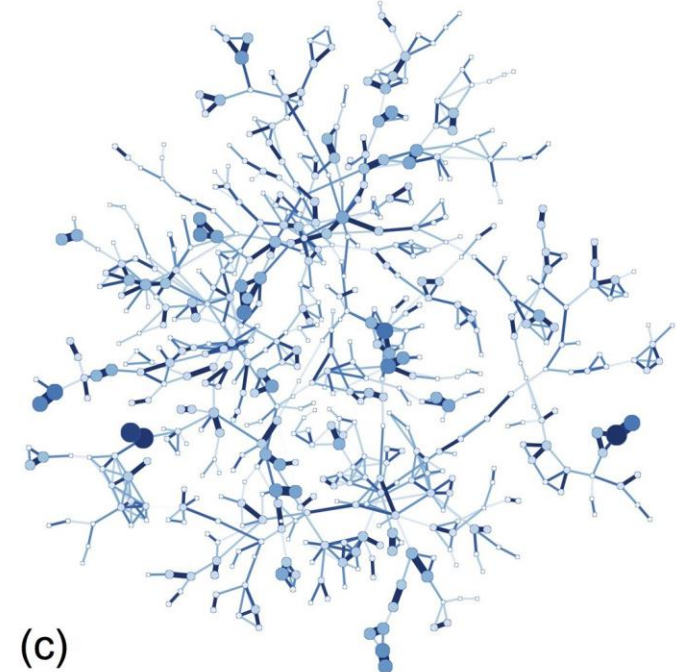
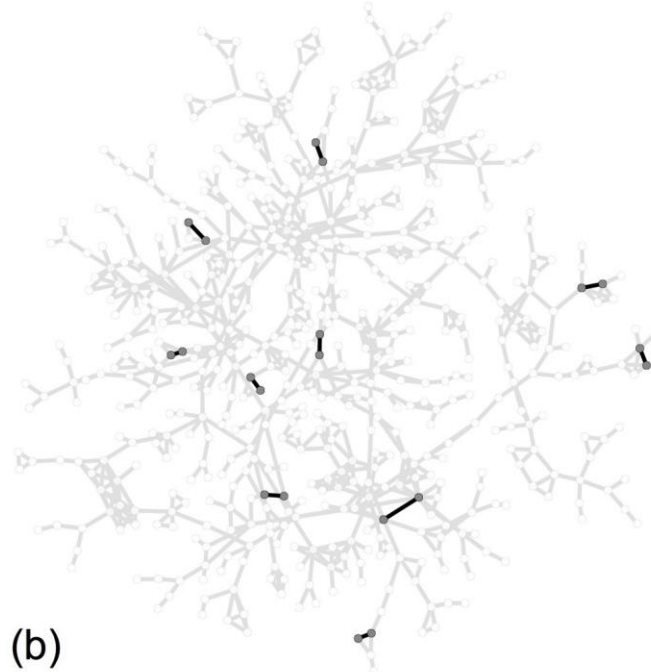
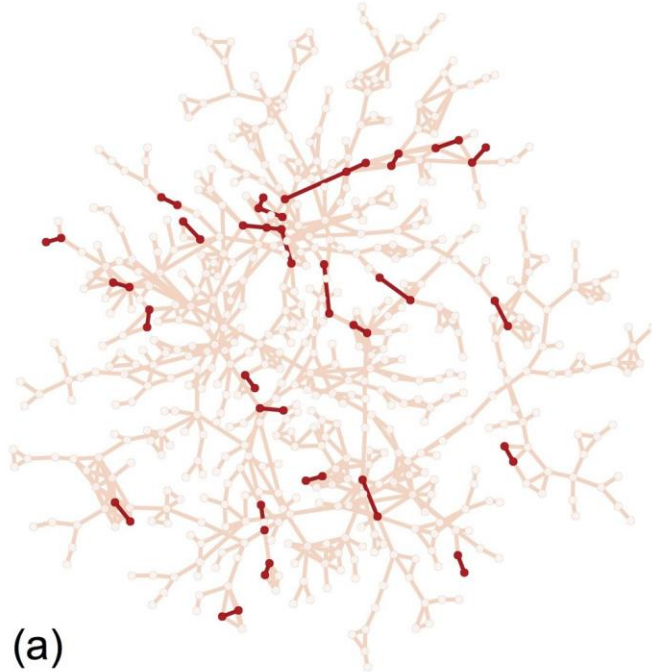
ER Random Network
Day 1



Speed of diffusion



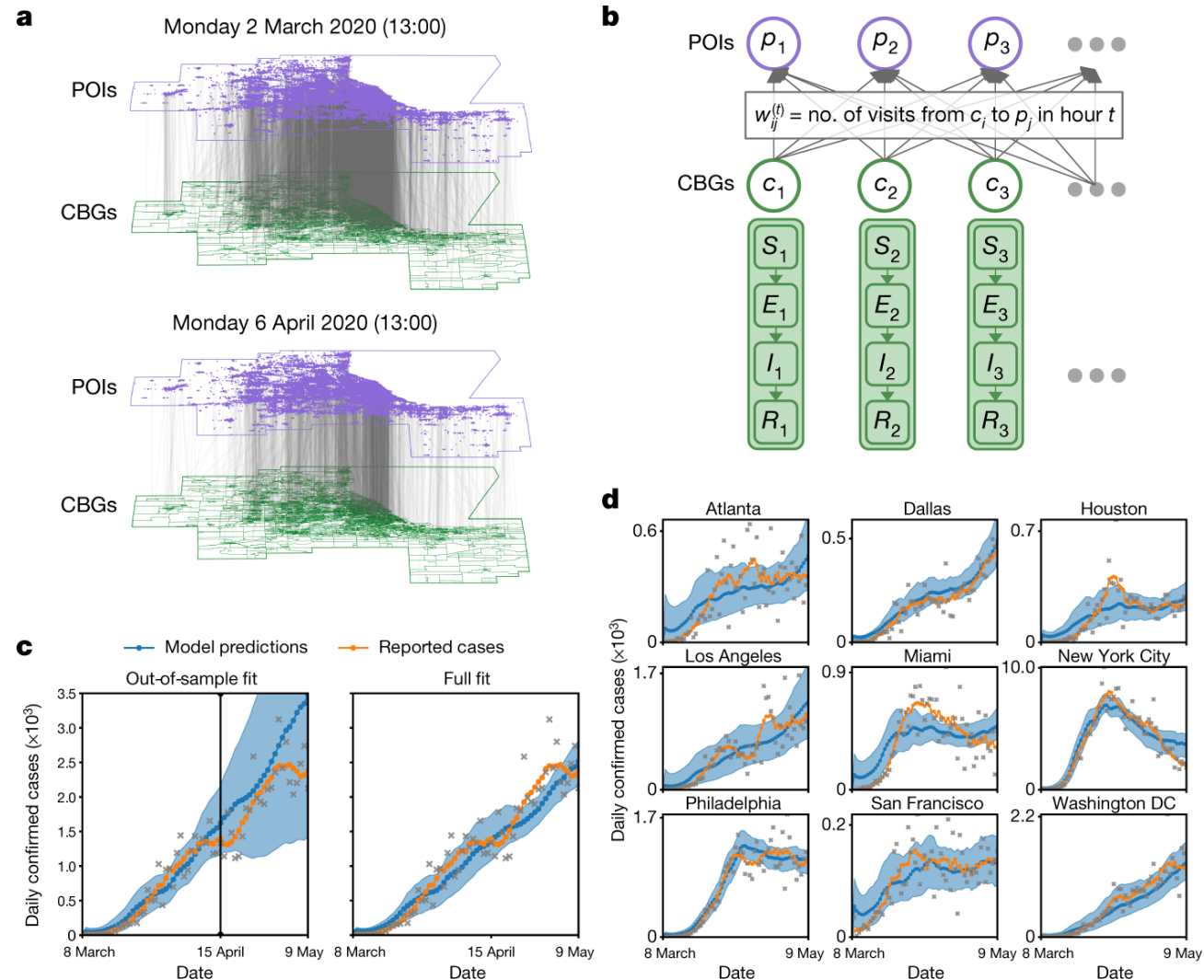
Dynamic networks



Time-varying edges slow down diffusion

- Karsai M, N Perra, A Vespignani (2014) SciRep
- Karsai M, M Kivelä, RK Pan, K Kaski, J Kertész, AL Barabási, J Saramäki Physical Review E 83 (2), 025102

Epidemiology and mobility: metapopulation models



Chang, S., Pierson, E., Koh, P.W. *et al.* Mobility network models of COVID-19 explain inequities and inform reopening. *Nature* **589**, 82–87 (2021).

I. 3. Simple versus Complex diffusion: spreading of viruses versus innovation

Threshold models of adoption

Shelling: segregation in cities

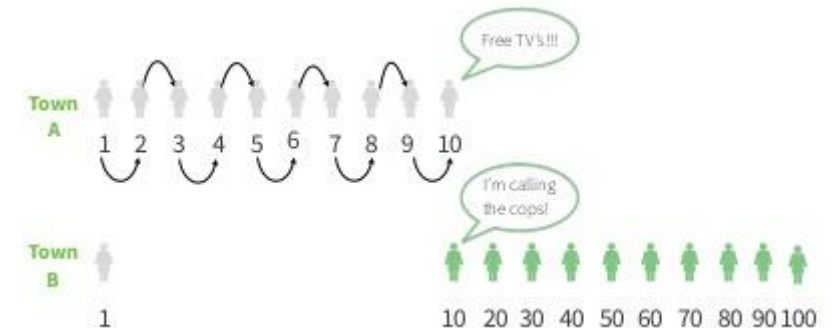
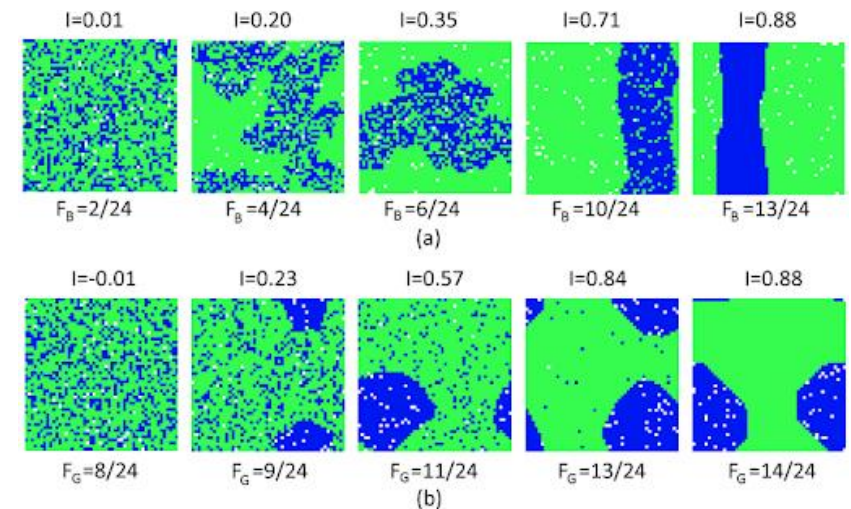
- Individual threshold in terms of ratio of tolerated different neighbors is heterogenous
- Low threshold individuals move away first that increases the probability of movement of high threshold individuals

Granovetter: behavior in a protest

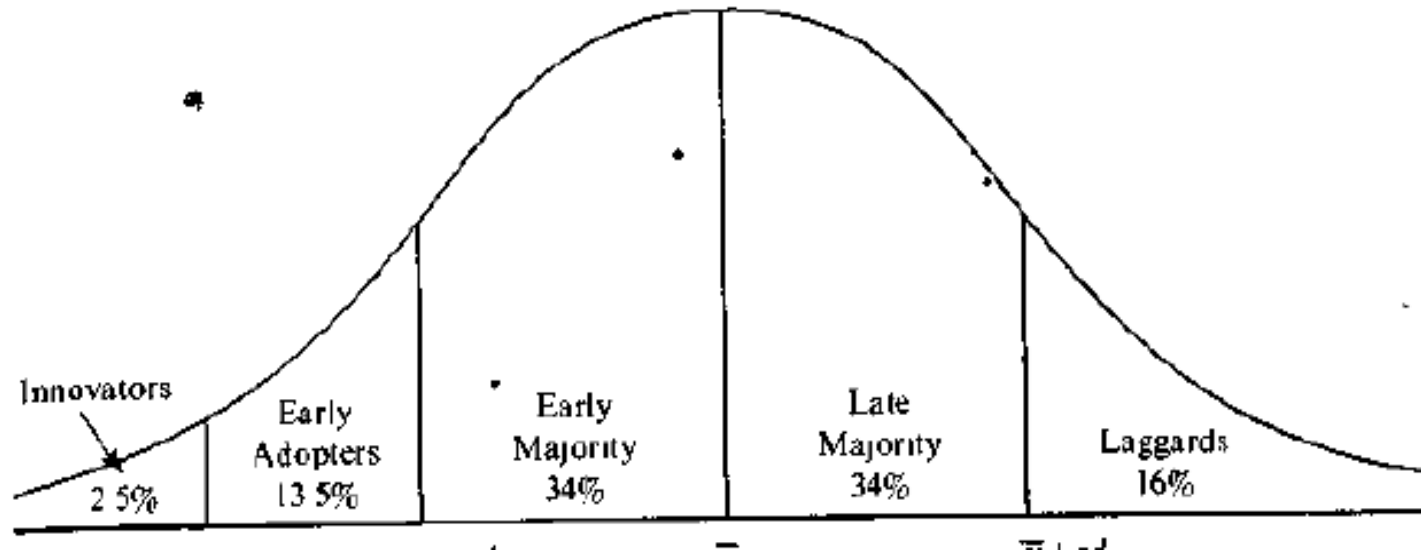
- Individual threshold: how many individuals is needed to start the activity before the individual will decide to start

Watts: cascading behavior in networks

- A behavior will diffuse only if the network structure follows the threshold sequence
- Innovators are connected to Early Adopters



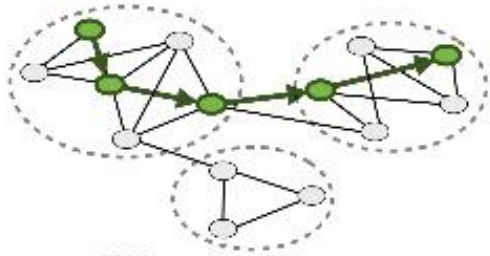
Adopter types over the life cycle



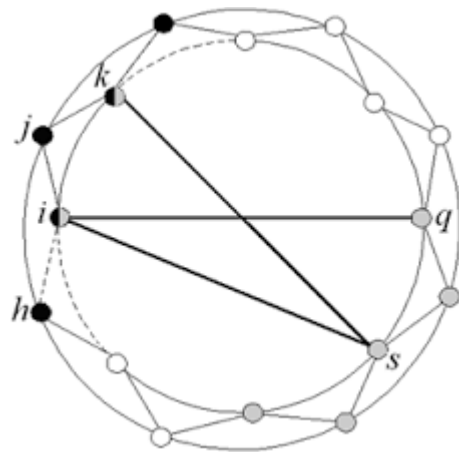
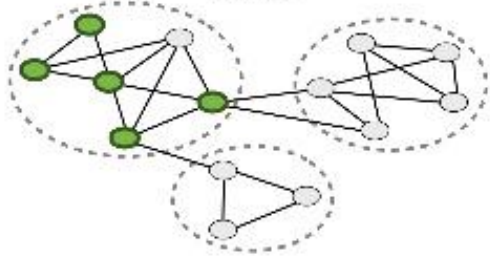
Forrás: Rogers (The Free Press, 1962),

Simple vs. complex contagion

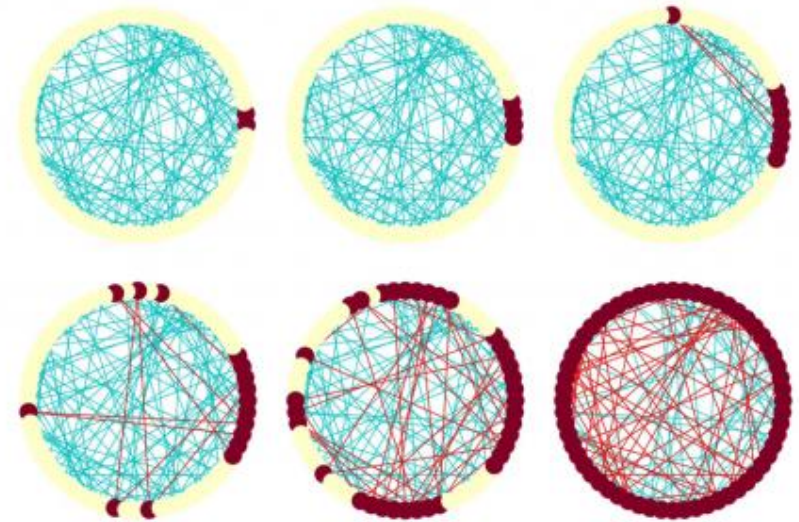
Simple contagion:
weak concentration



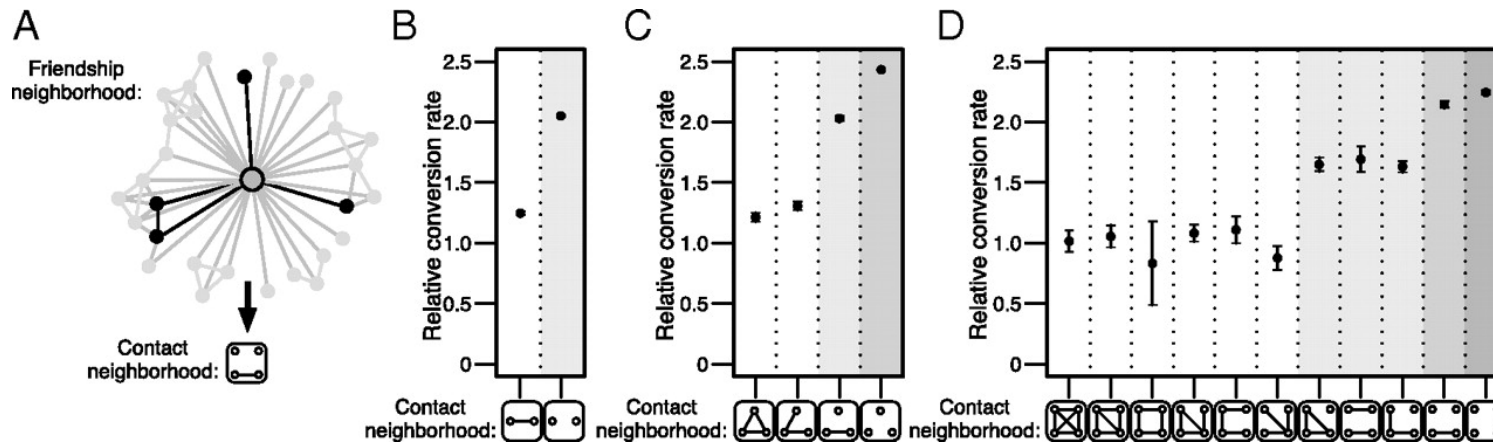
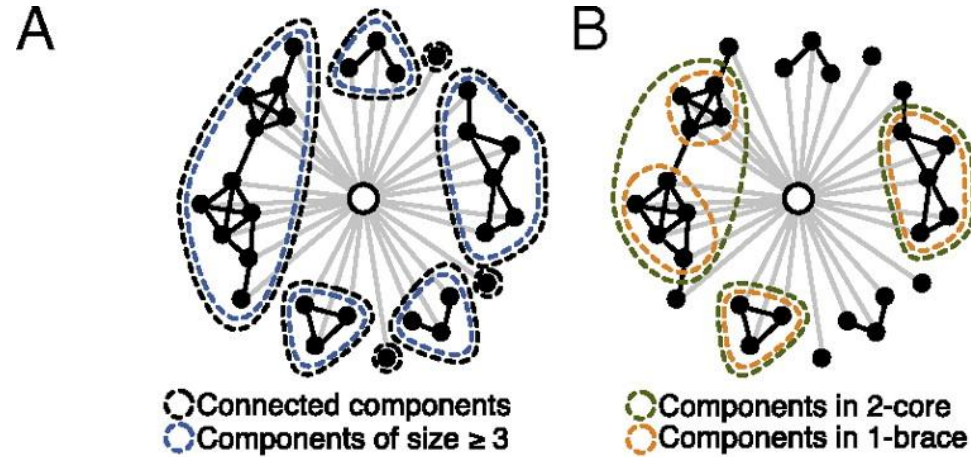
Complex contagion:
strong concentration of
communication inside
communities



- Simple: infection depends on dyadic probability
- Complex: adoption is a process of convincing-decision and depends on fraction/number of adopting friends
- Innovation diffuses through complex contagion.
- Small-world networks: high clustering speeds diffusion up, bridges slow it down.

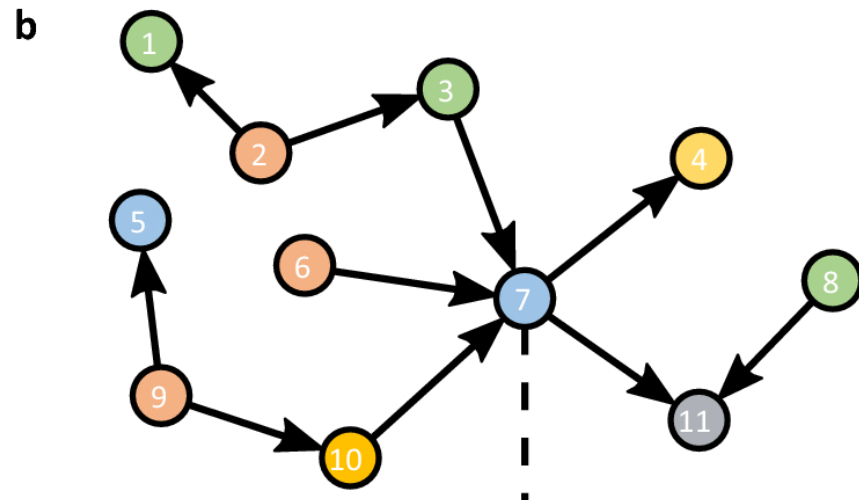
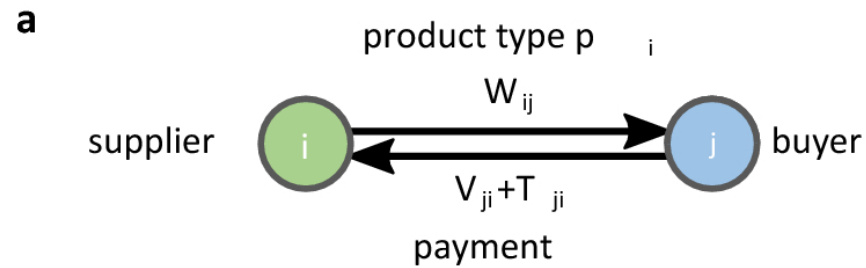


Role of communities: adoption probability increases after adoption in diverse communities

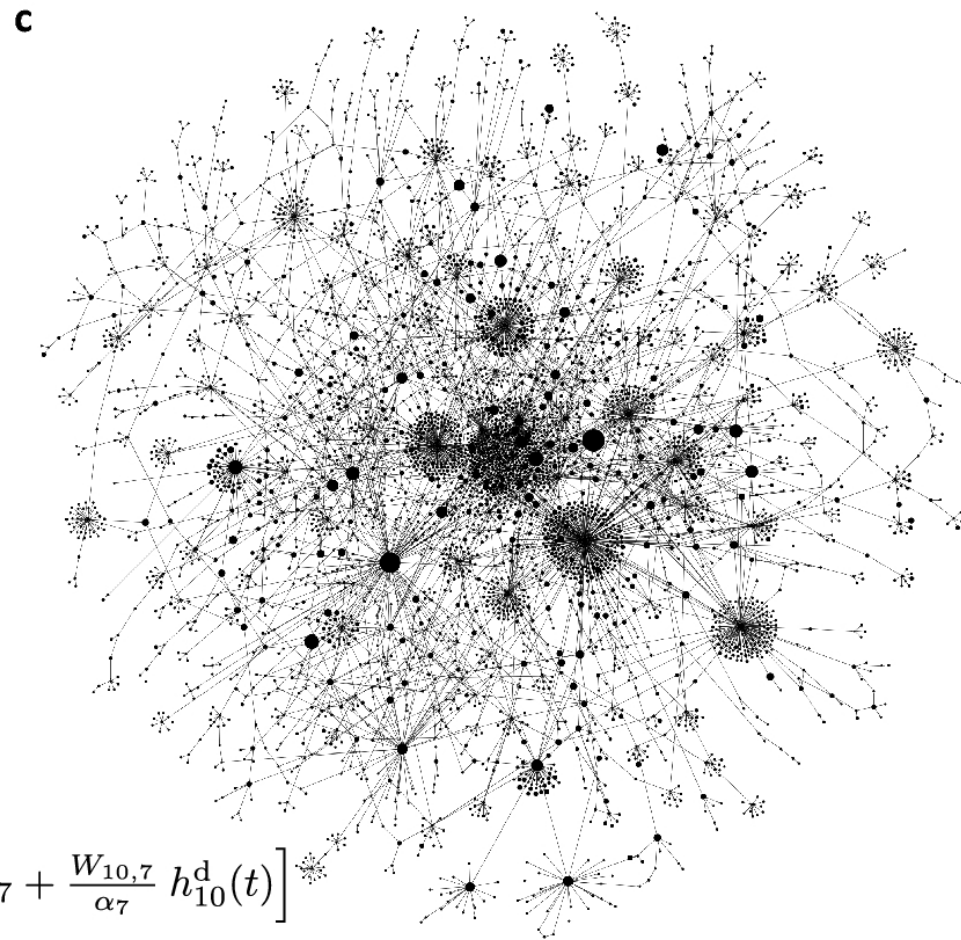


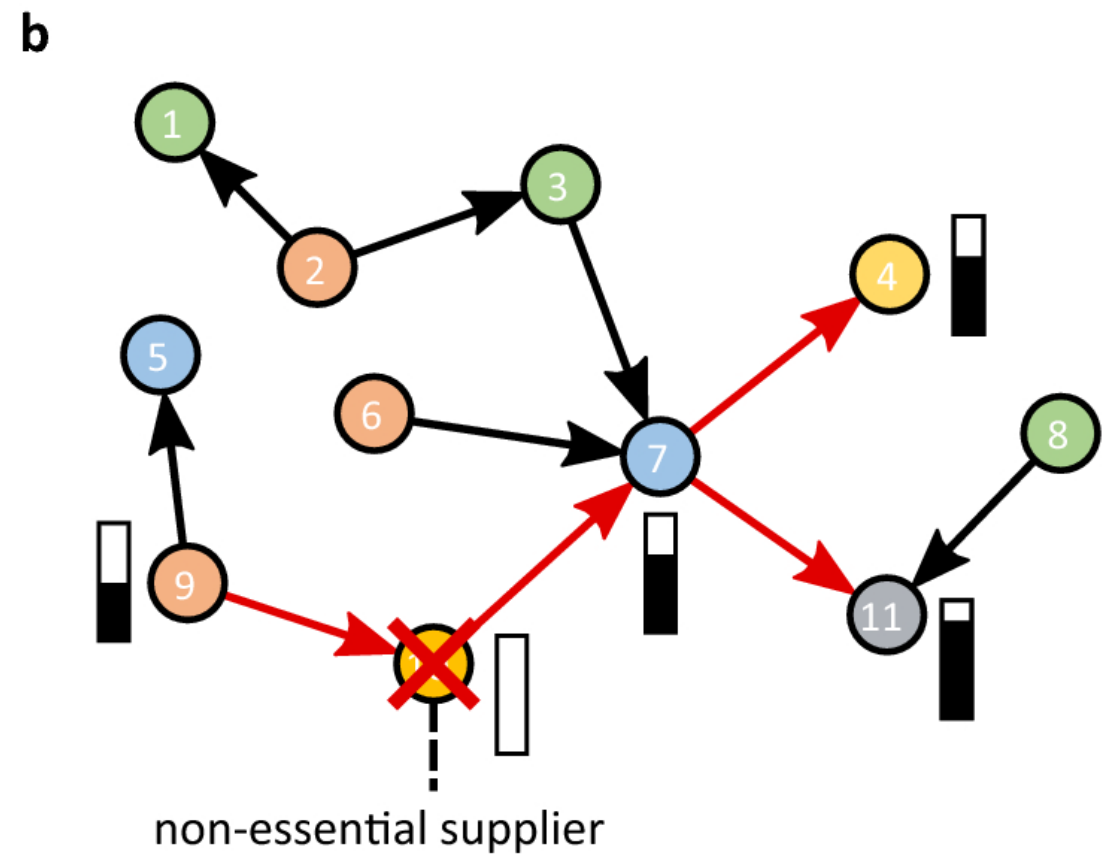
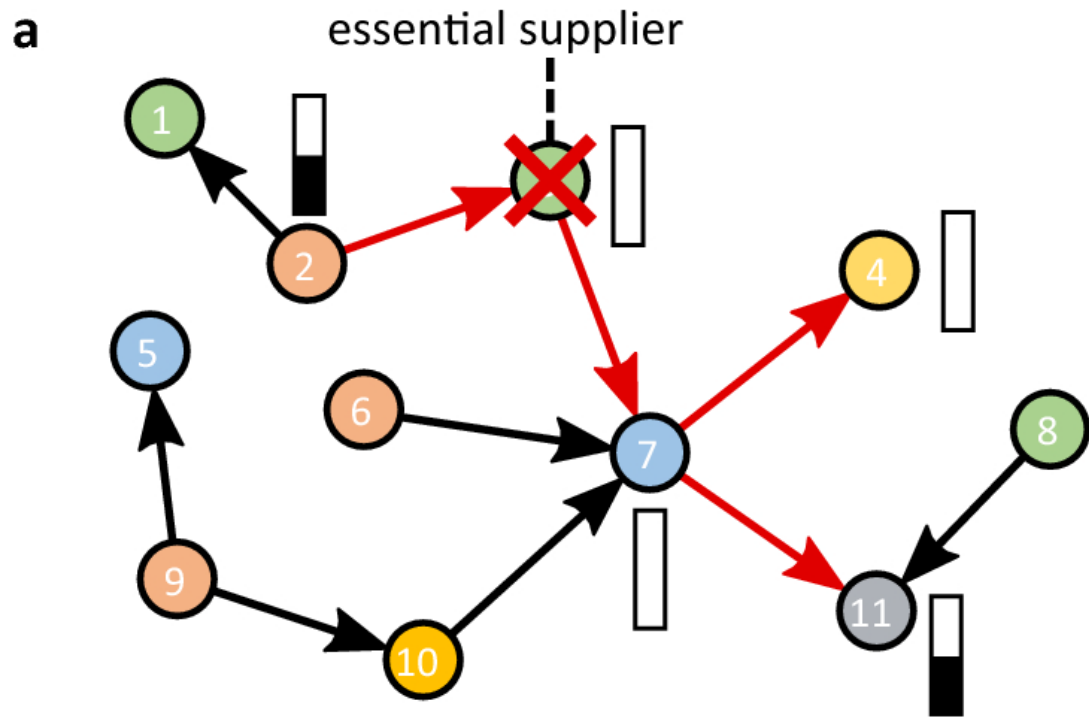
II. 1 Shock propagation

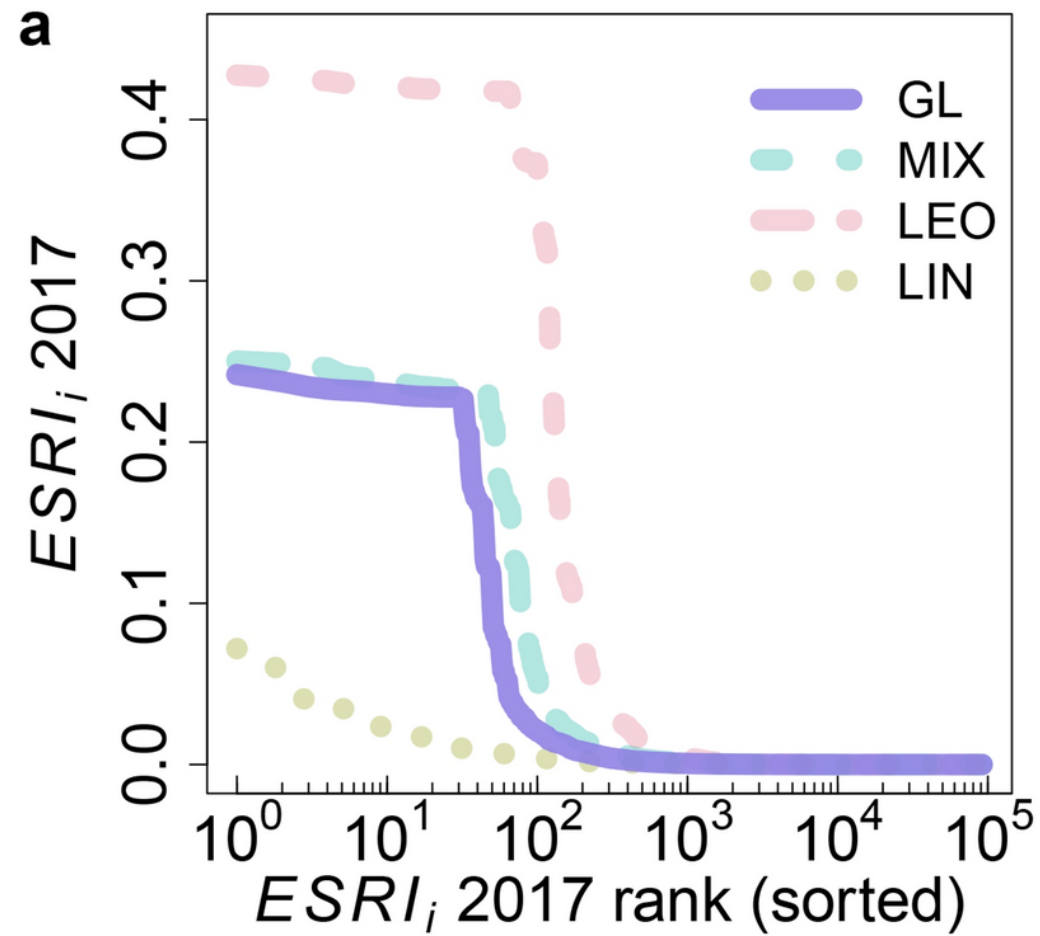
Shock propagation in supply chains



$$x_7(t+1) = \min \left[\min \left[\frac{W_{3,7}}{\alpha_{7,1}} h_3^d(t), \frac{W_{6,7}}{\alpha_{7,2}} h_6^d(t) \right], \beta_7 + \frac{W_{10,7}}{\alpha_7} h_{10}^d(t) \right]$$







Estimating the loss of economic predictability from aggregating firm-level production networks

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^fSanta Fe Institute, Santa Fe, NM 85701, USA

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Edited By: Taylor Jaworski

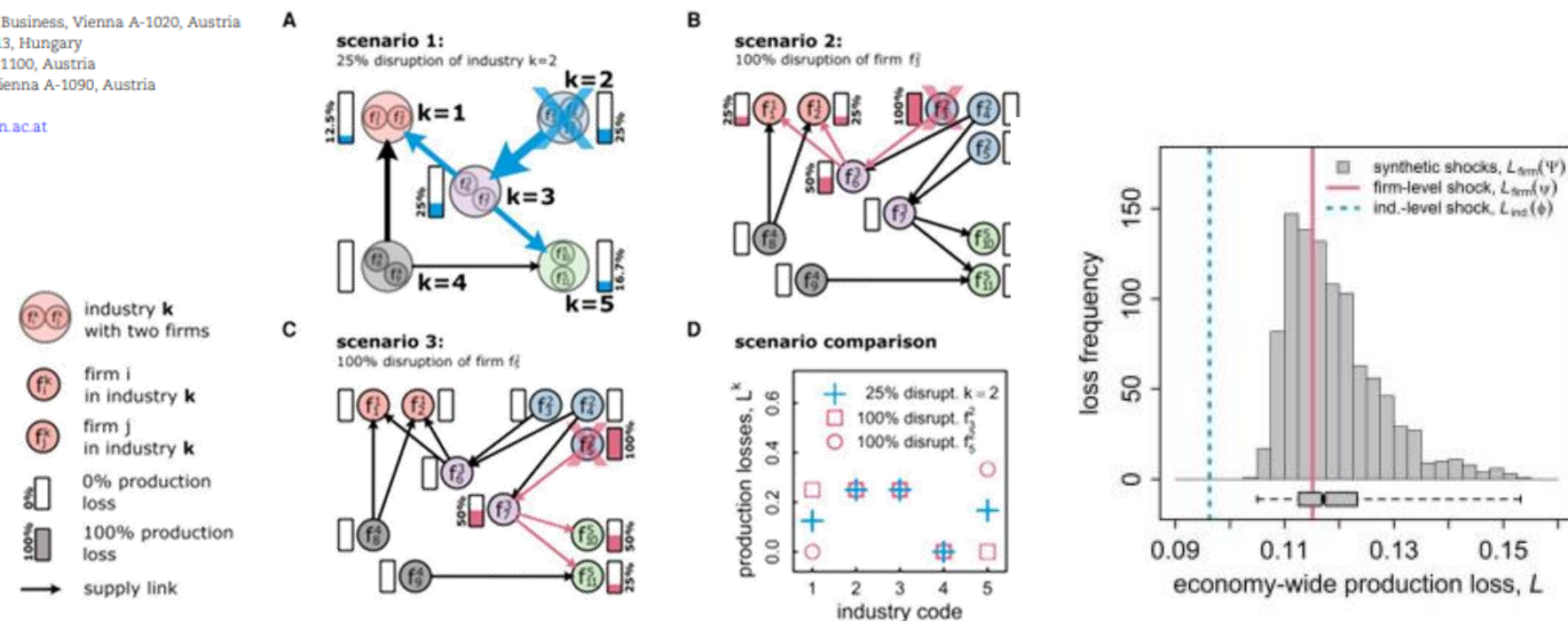
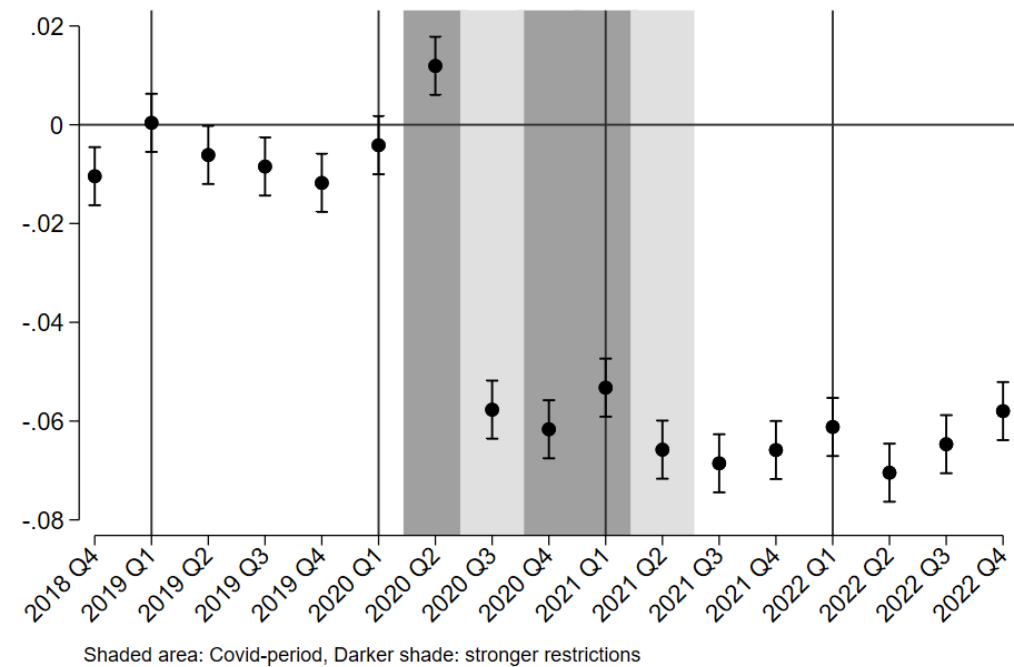
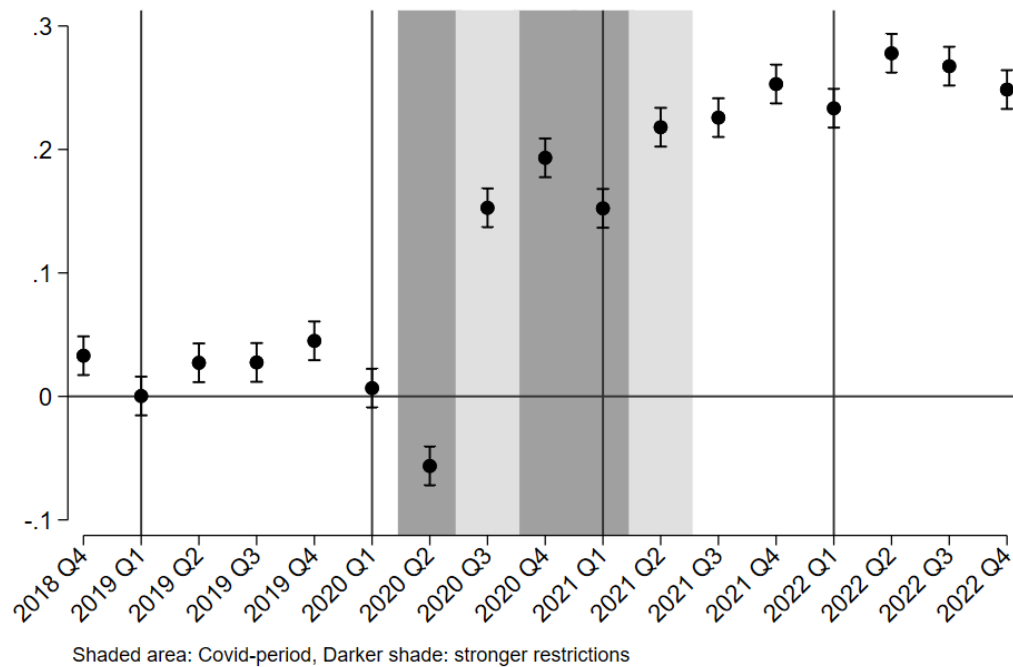


Fig. 1. Schematic demonstration of how errors in production loss estimates originate from aggregating a FPN to the IPN. A) Scenario 1, shock propagation on the IPN in response to a 25% initial disruption of industry 2 (blue X) resulting in a 25% production loss (blue bar marks 25% reduction). The disruption of sector 2 can originate from various combinations of shocks to firms f_1^2 , f_2^2 , and f_3^2 . The initial shock spreads downstream to sector 3, leading to a 25% production loss, and further to sector 1 (12.5% loss) and sector 5 (16.7% loss). B) Scenario 2, shock propagation on the FPN in response to a 100% disruption of firm 3 (red X, red bar). The disruption propagates downstream (red edge) to firm 6 (50% production loss), and further to firms 1 and 2 (25% loss). Other nodes are not affected (0% loss, empty bars). C) Scenario 3, shock propagation on the FPN in response to a 100% disruption of firm 5 (red X, red bar), resulting in a 0% production level. The disruption propagates downstream (red edge) to firm 7 (50% loss) and to firms 10 (50% loss) and 11 (25% loss). Other nodes are not affected. D) Comparison of industry-specific production losses, L^x (y-axis) for industries 1, 2, 3, 4, and 5 (x-axis), in response to the aggregated 25% disruption of sector 2 (blue “+”) and the two 100% firm-level shocks, of firms 3 and 5 (red squares and circles). Note that aggregating both firm-level shock scenarios leads to the same 25% shock of industry 2. The production losses of industries 2, 3, and 4 are 0.25, 0.25, and 0, respectively, for all three cascades the symbols “+”, circle, and square overlap. However, the output losses of sectors 1 and 5 are remarkably different for the three different shocks (symbols do not overlap). FPN-losses are seen to vary from 0 to 0.25 for sector 1, and from 0 to 0.33 for sector 5, whereas the IPN losses are the same for both firm-level failure scenarios: 0.125 for sector 1, and 0.167 for sector 5. Remarkably, the IPN-loss estimates deviates from the FPN-loss estimates by about 100%.

Firms actively react to supply chain shocks to mitigate risks (Békés, Bisztray, Koren, Lengyel, Neubrandt)



Evolution and determinants of firm-level systemic risk in local production networks

Anna Mancini^{1,2,*}, Balázs Lengyel^{3,4,5}, Riccardo Di Clemente^{6,7}, and Giulio Cimini^{1,2,8}

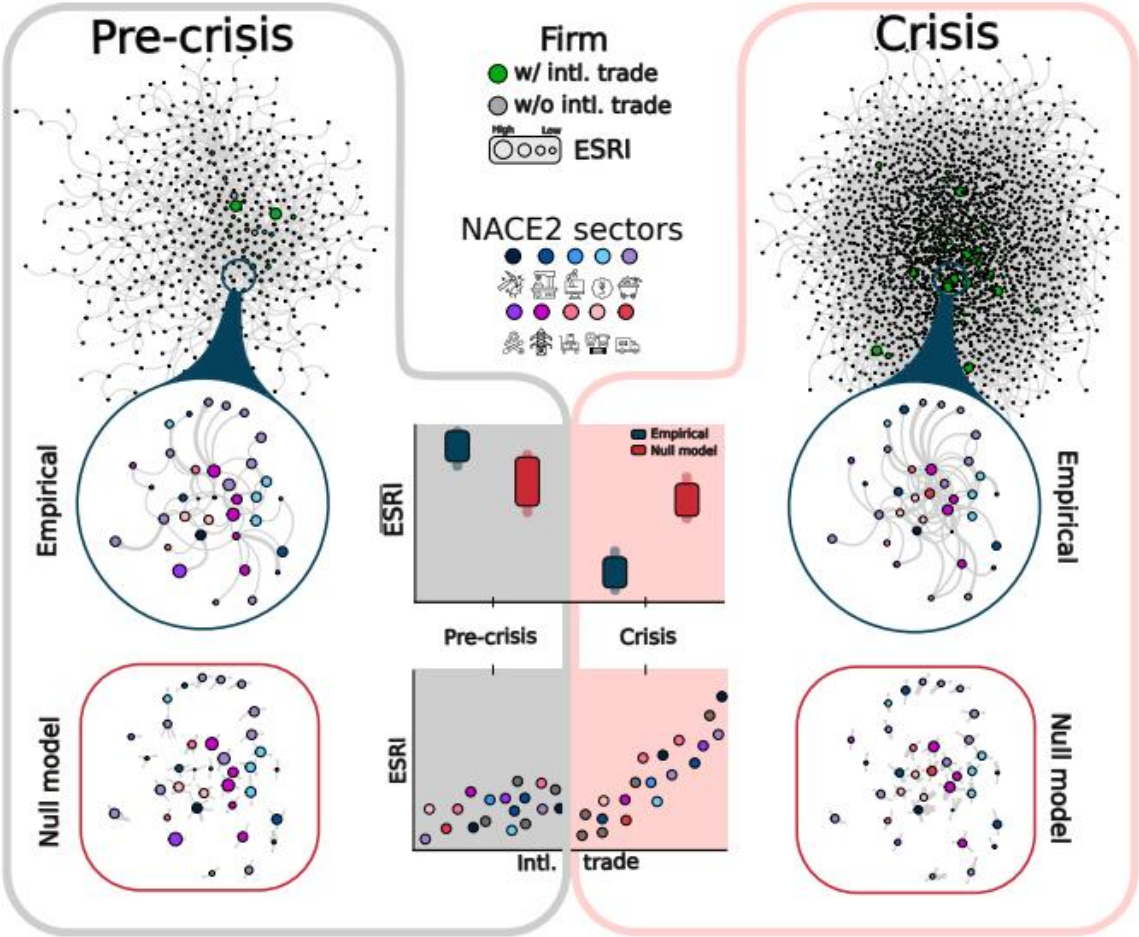
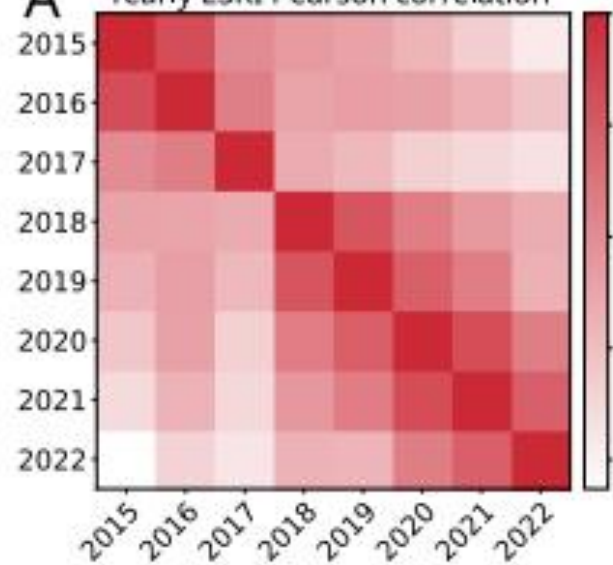
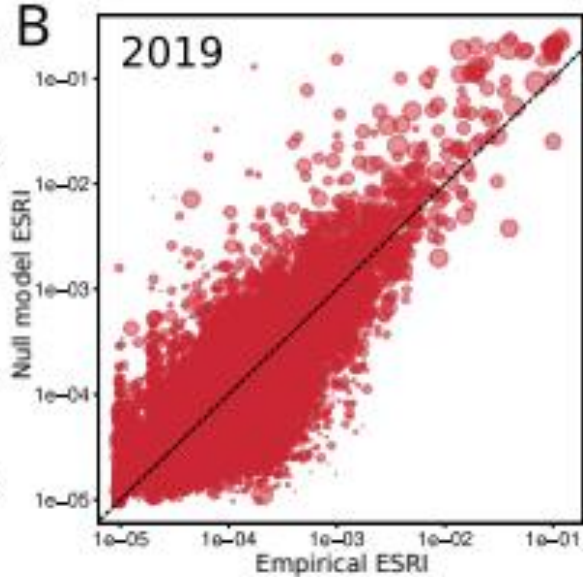


Figure 1. Empirical and null model networks are compared before and during the COVID-19 crisis, to study the evolution of economic systemic risk and understand the role of international trade. We start with temporal snapshots of the empirical production network. The top of the figure shows the network of a sample of the same 1000 nodes, before and during the crisis. Firms are colored according to the presence (green) or absence (gray) of international trade ties, while nodes' size is proportional to firms' ESRI value. We then focus on firms with the highest systemic risk values, highlighted with a circular magnification from the empirical networks, where the nodes' color represents the firms' industrial (NACE2) sector. Finally, the squares in the third row represent the null models for the same set of risky firms, where we show only the constrained quantities (for each firm, its supplies from different sectors and total sales). Our framework thus allows studying the temporal evolution of systemic risk, how it deviates from the predictions of the null model, and which characteristics of the firms, such as the presence of international trade, are linked to their risk value (as shown by the sample analysis plots in the middle of the figure).

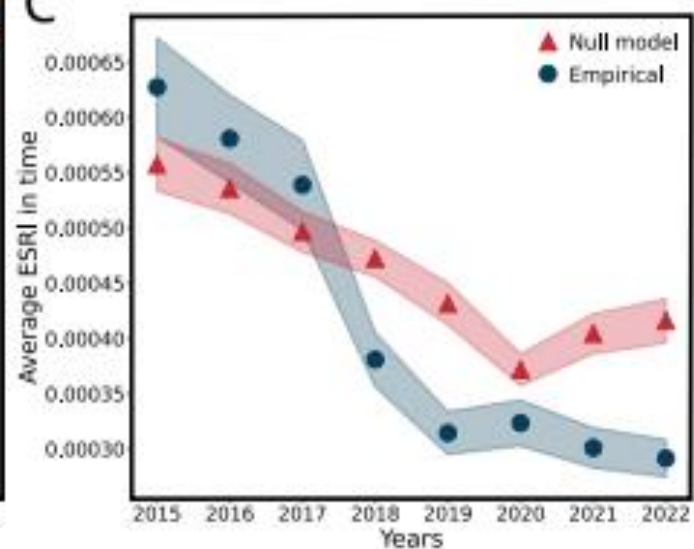
A Yearly ESRI Pearson correlation



B

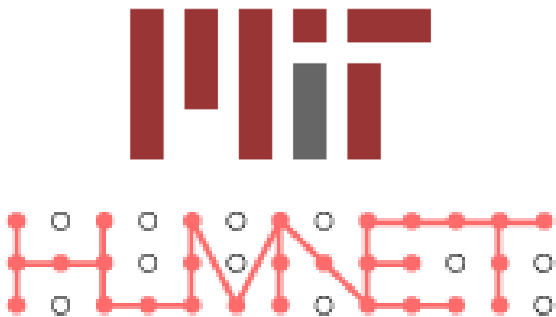


C



The role of geography in the complex diffusion of innovations

Lengyel Balázs



Bokányi Eszter
KRTK



Riccardo di Clemente UCL

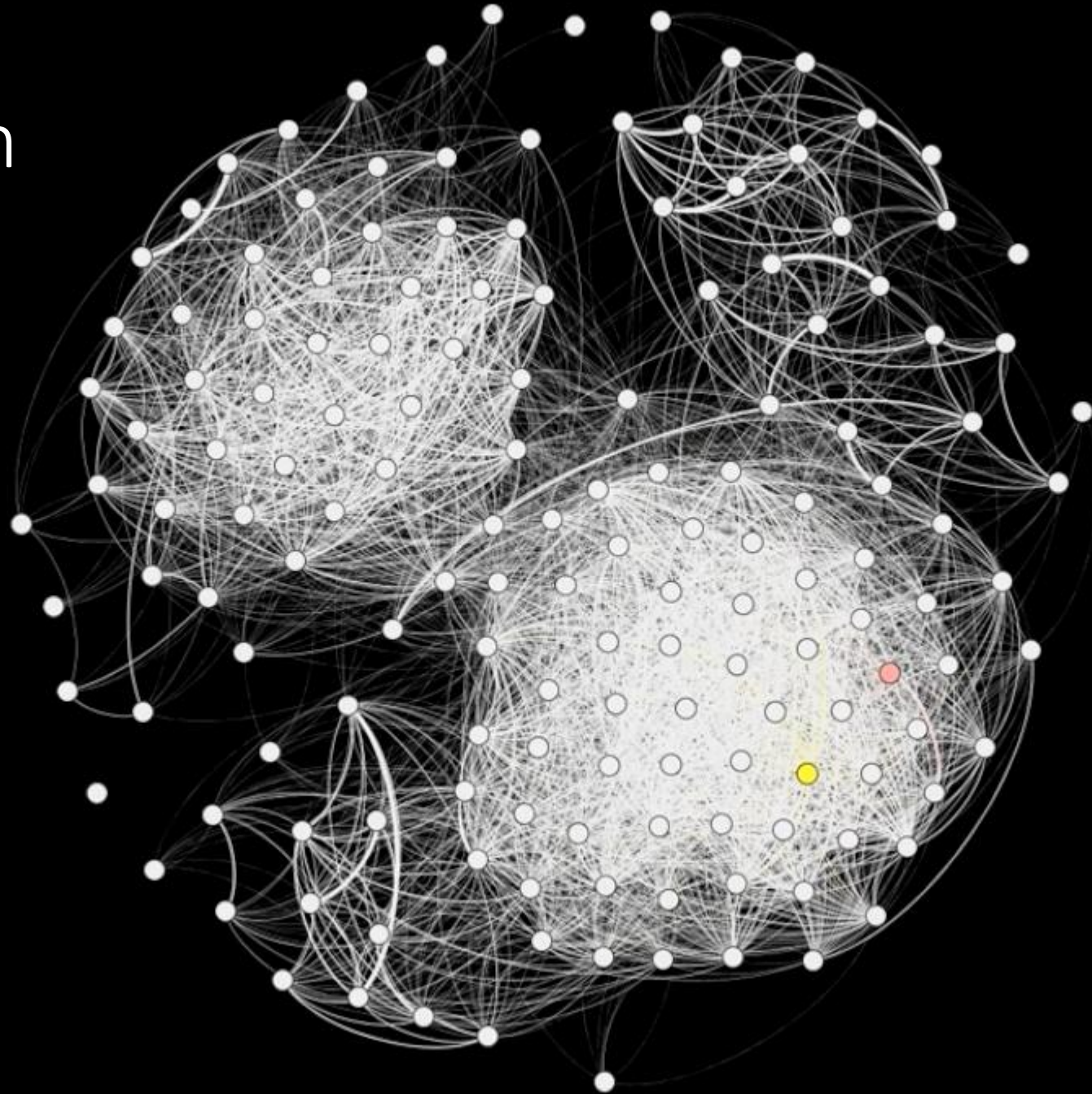


János Kertész CEU



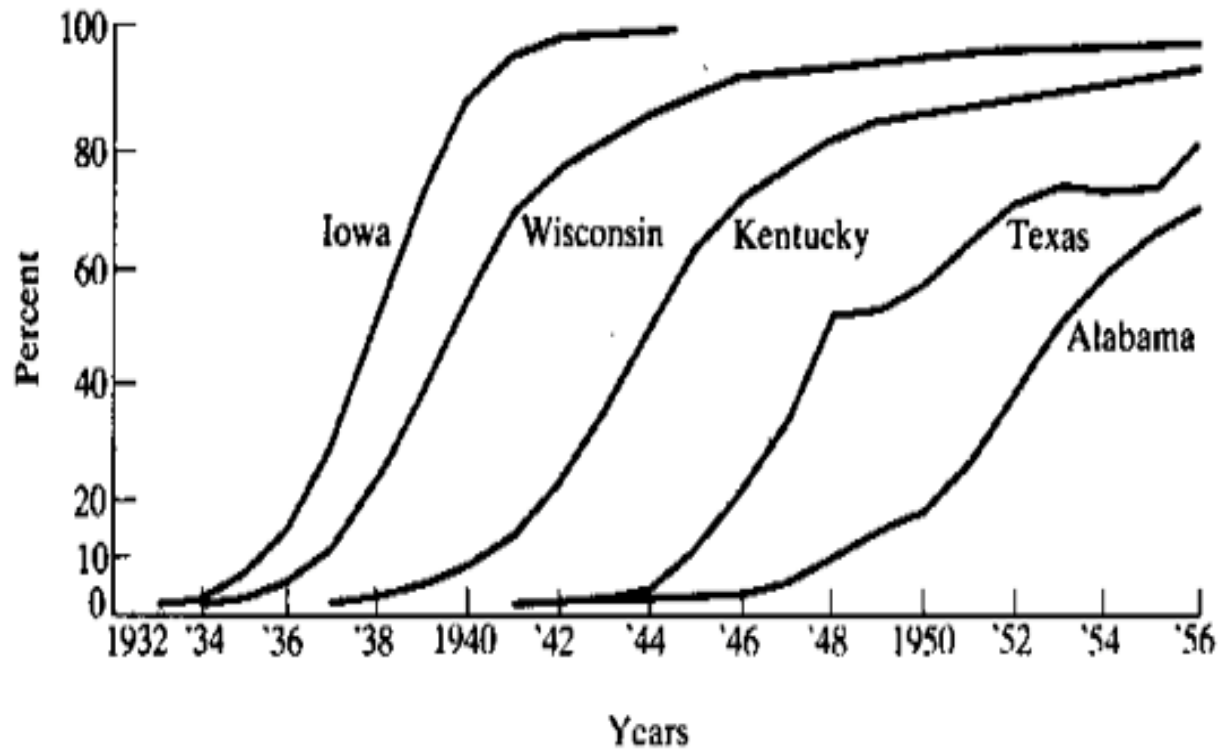
Marta González
UC Berkeley

Diffusion in networks



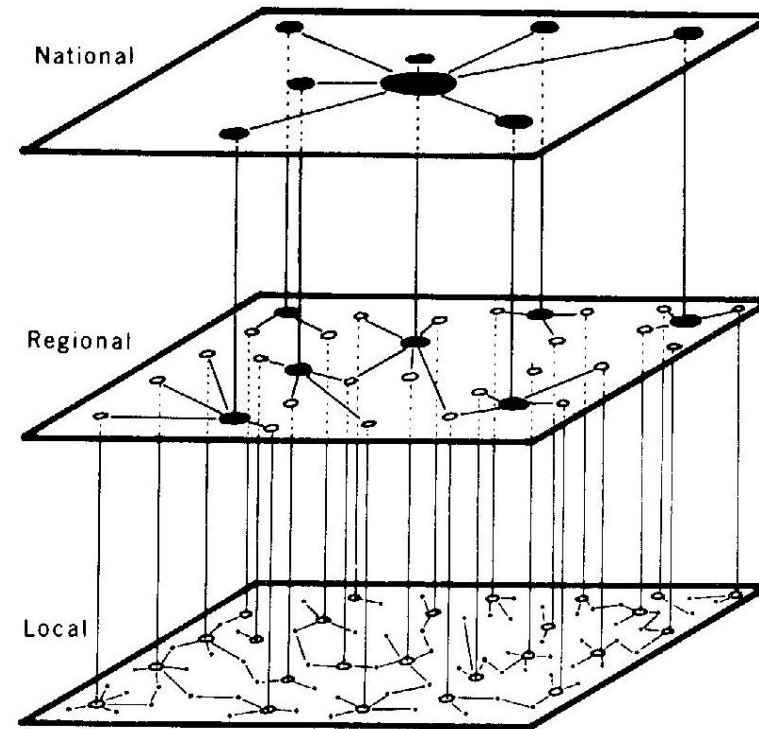
Spatial diffusion

Innovation gets quicker to proximate places.

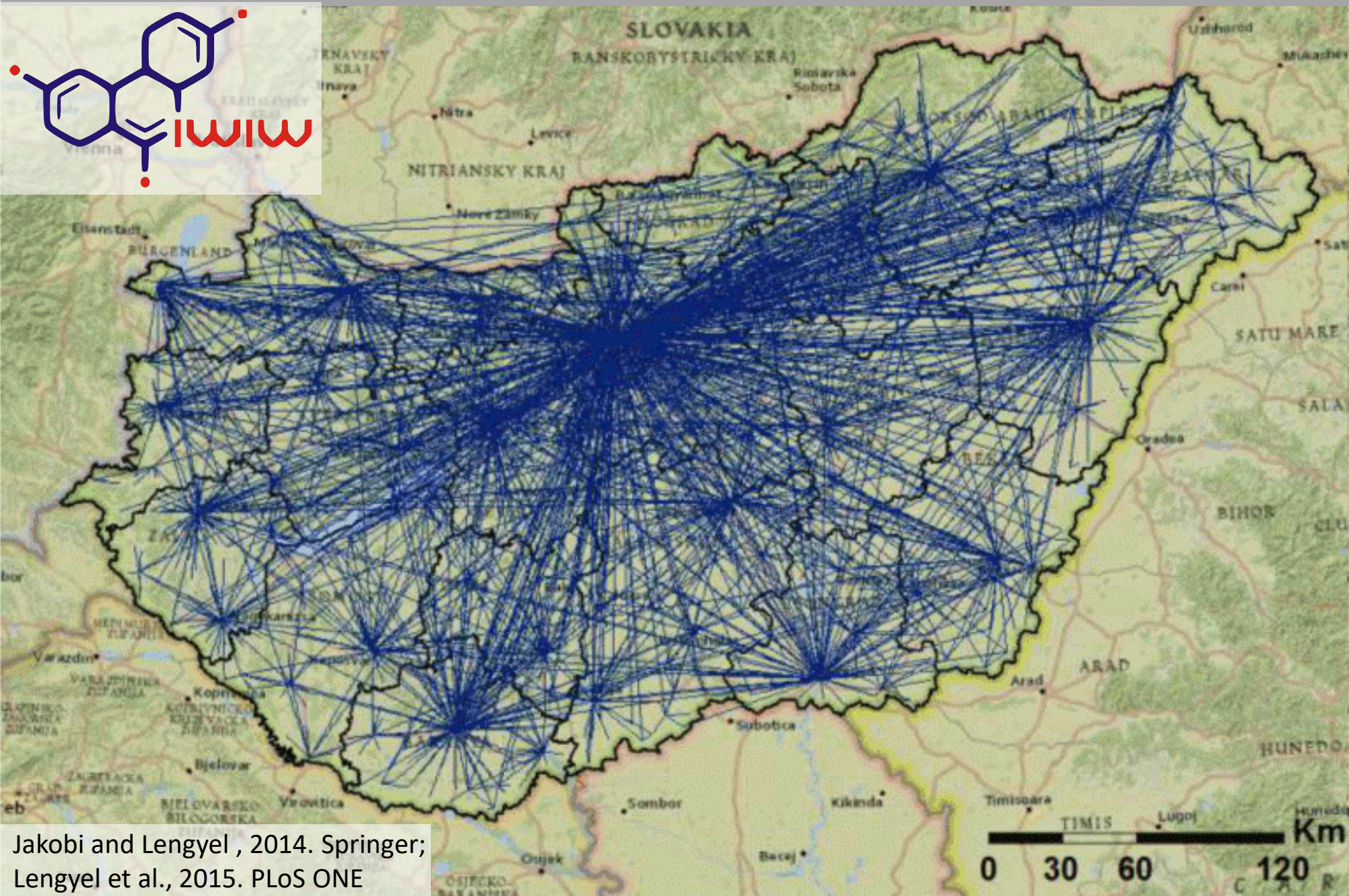
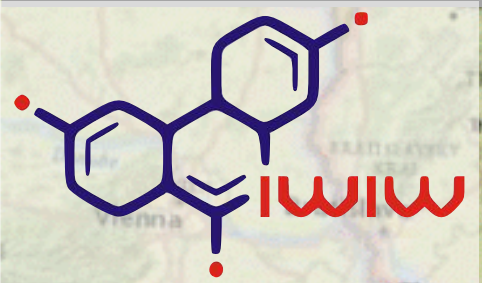


Griliches (1957, Econometrica)

Innovation spreads from big to medium cities and then to small towns.



Haegerstrand (1953, Chicago UP)

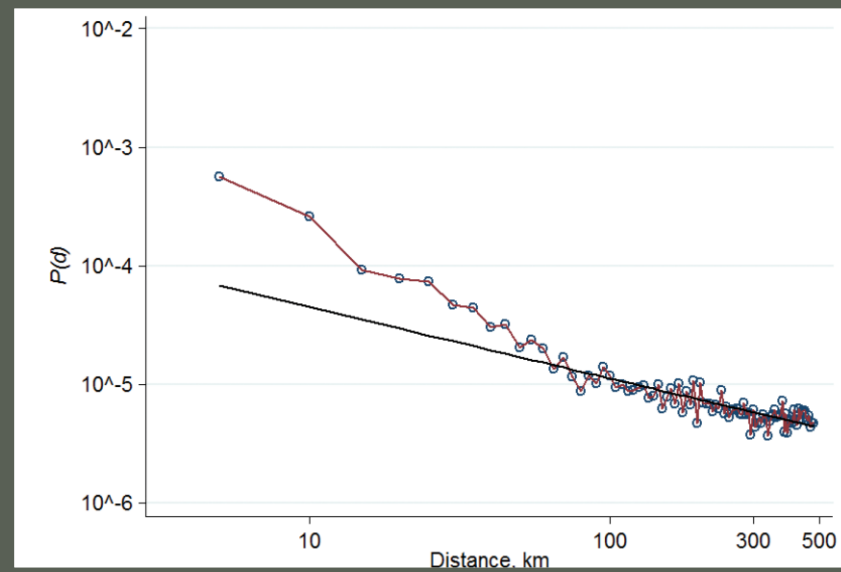
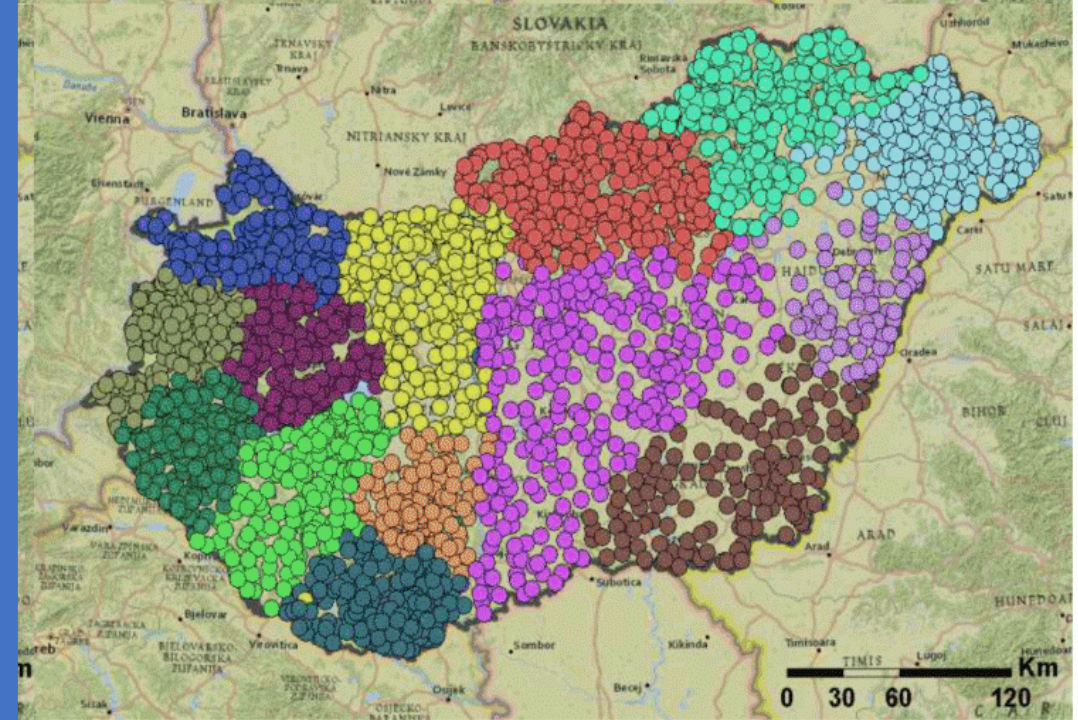
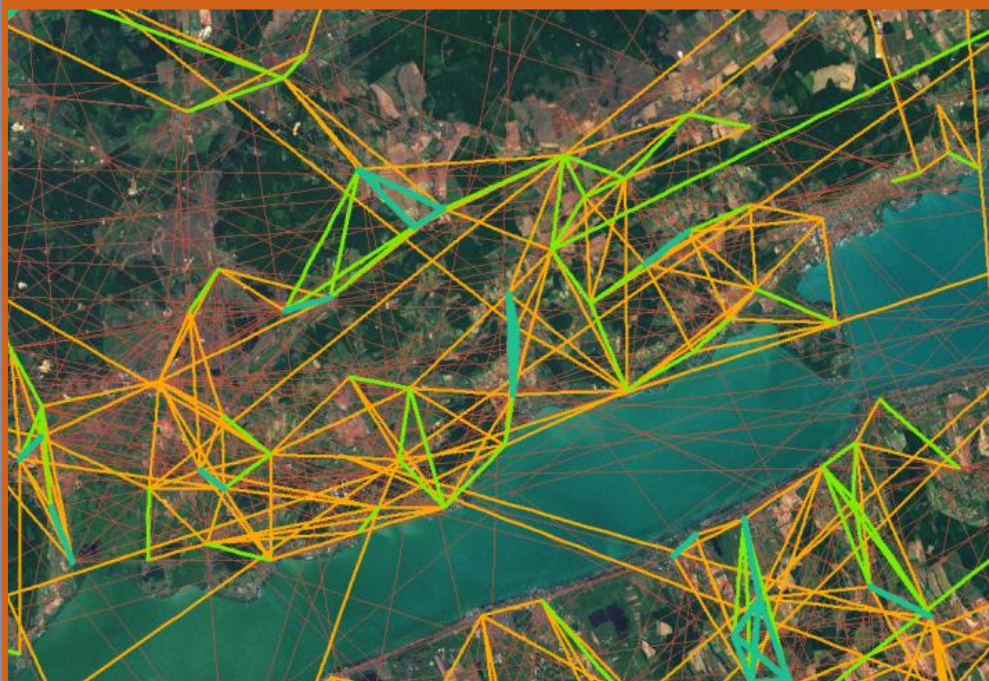
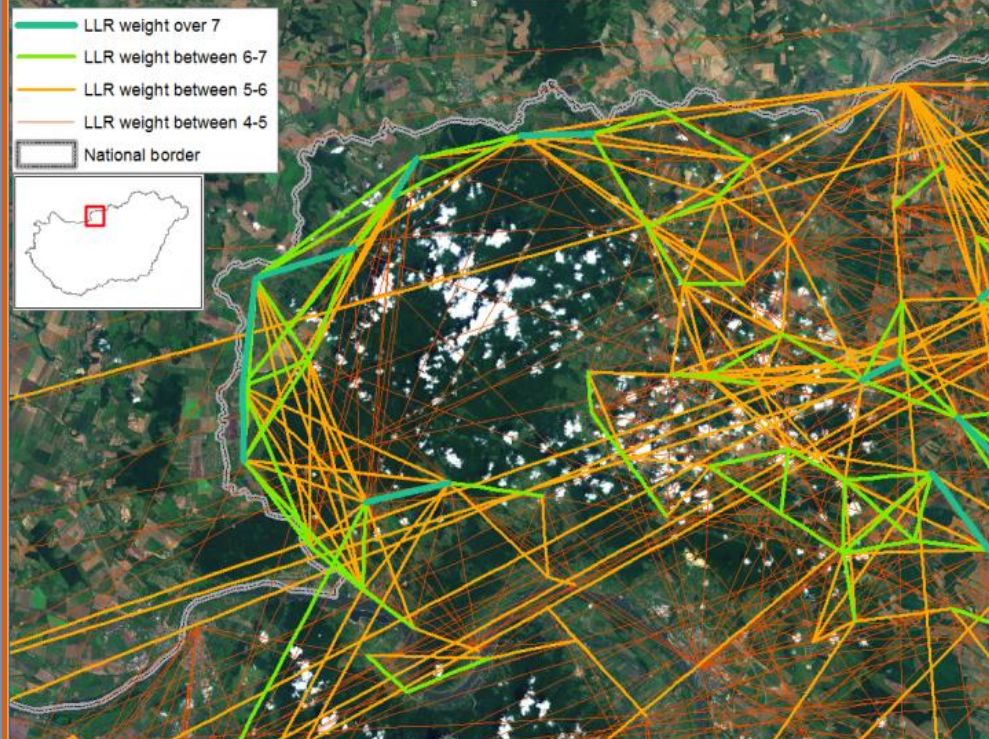


Full life cycle
of a social
media
platform

Users

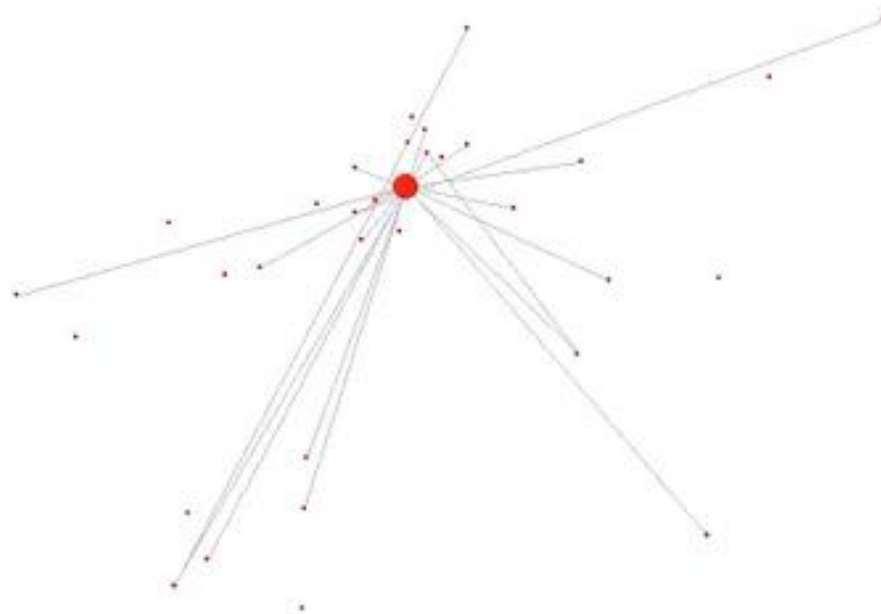
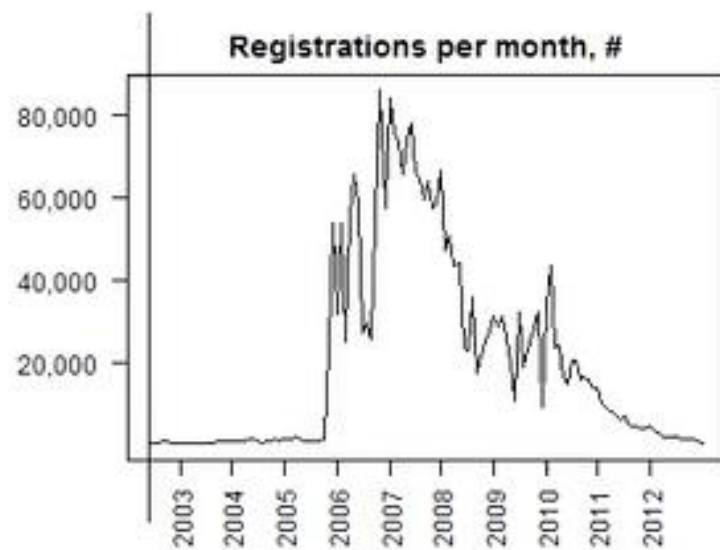
- **Location** (self-reported),
- **Date of registration,**
- **Date of last login,**
- **ID of friends,**
- **ID of invitor**

Geographies of the network



Registered users, #

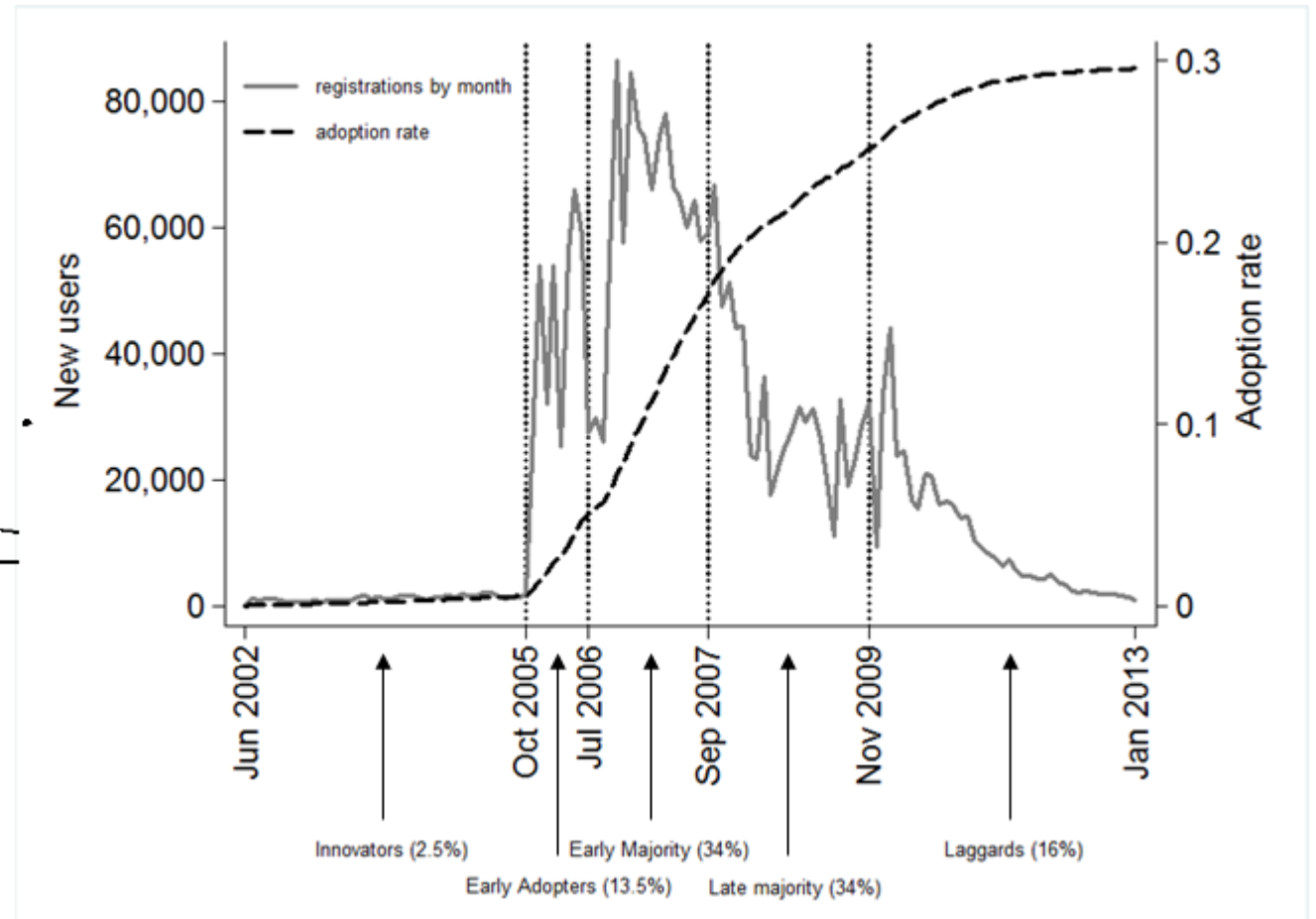
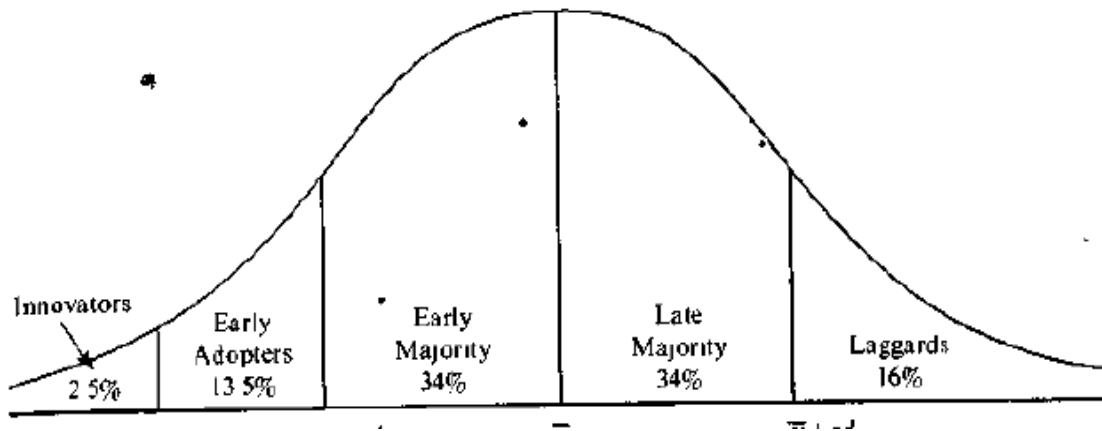
- 10
- 100
- 1000
- 10000
- 100000



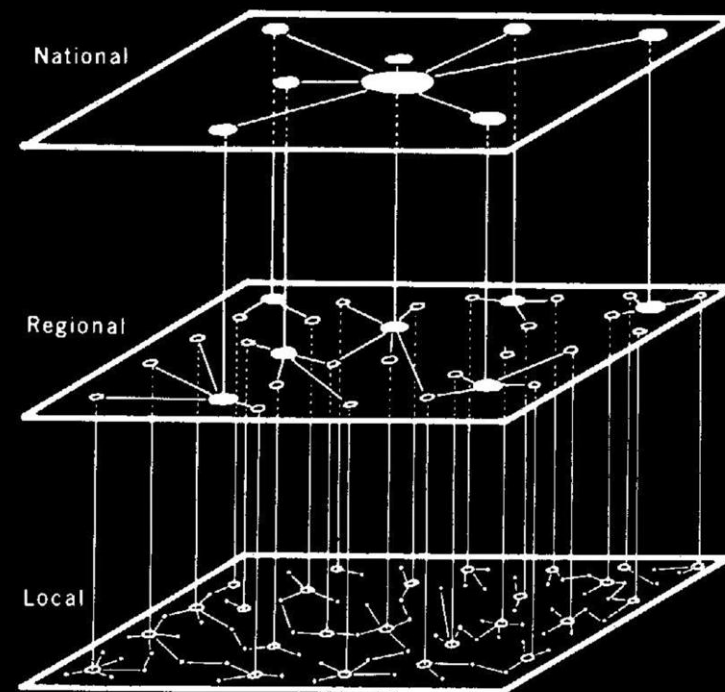
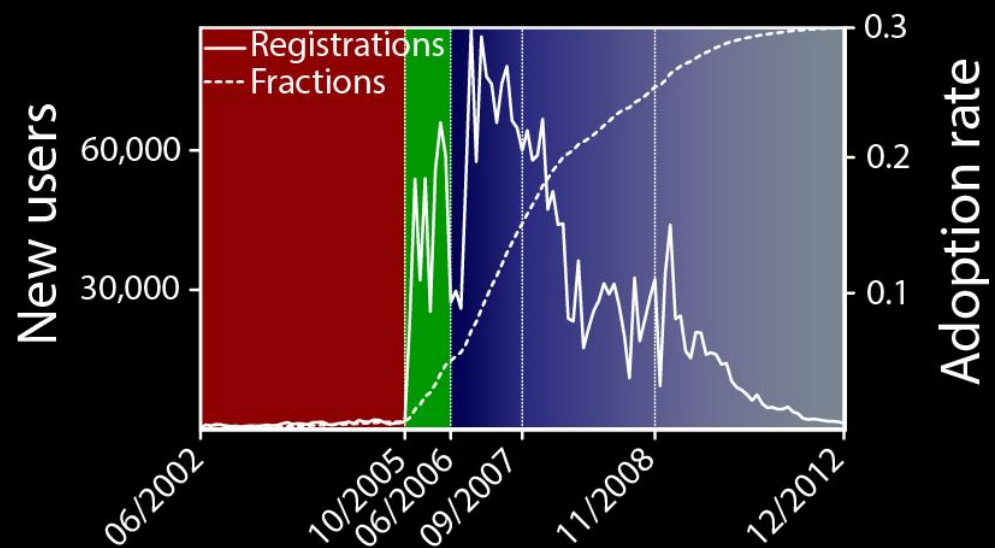
Active users, %



The iWiW life cycle by Rogers categories



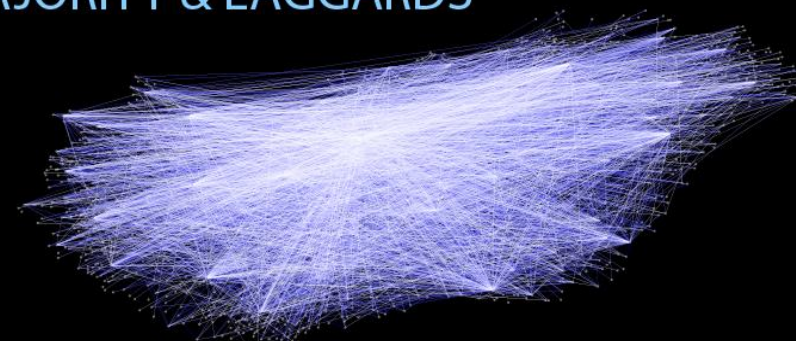
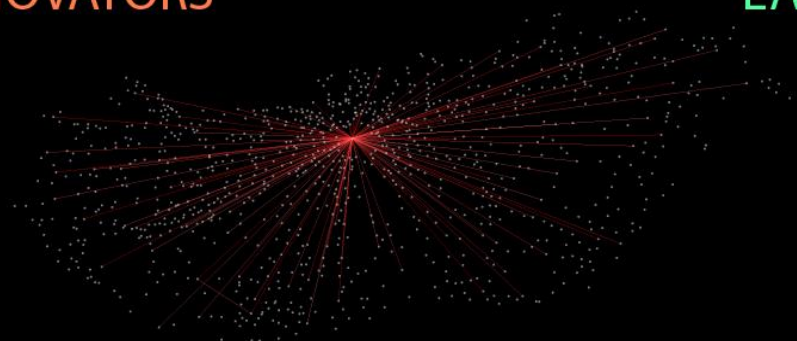
Forrás: Rogers (The Free Press, 1962),



INNOVATORS

EARLY ADOPTERS

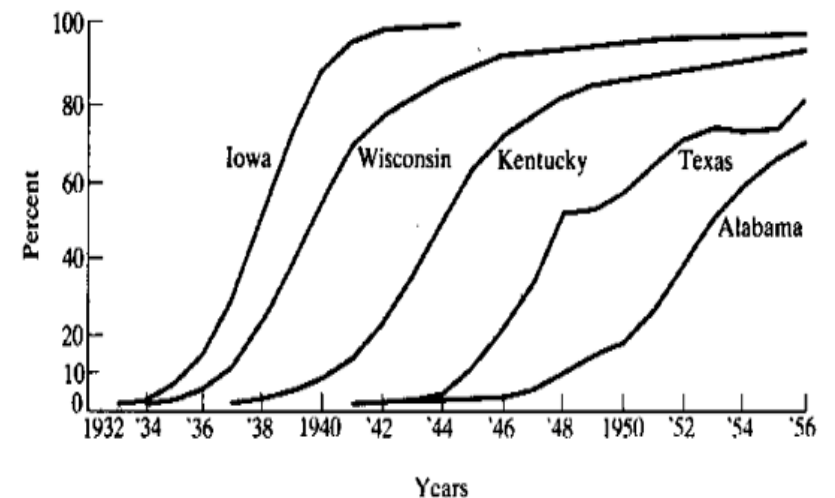
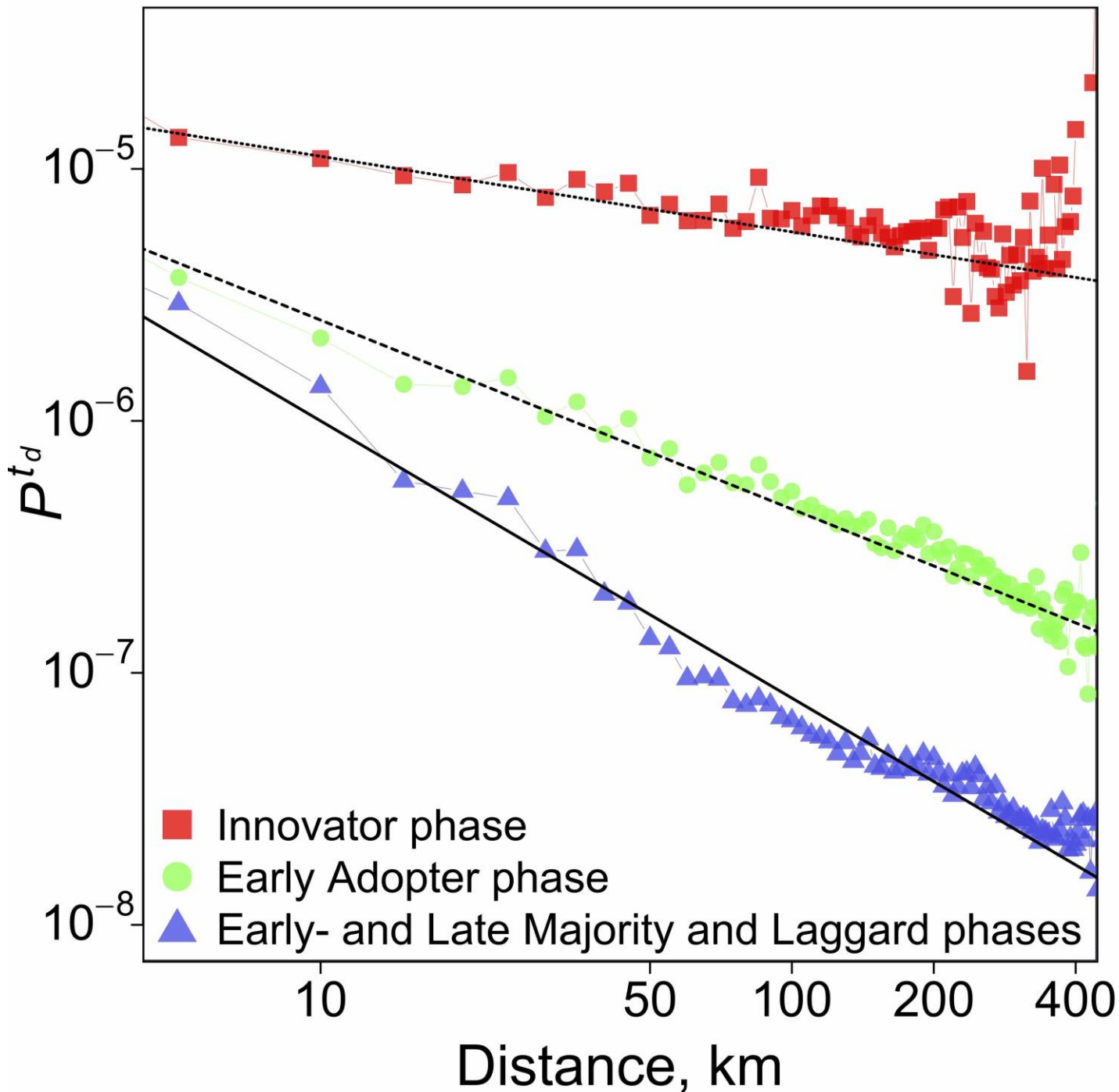
MAJORITY & LAGGARDS

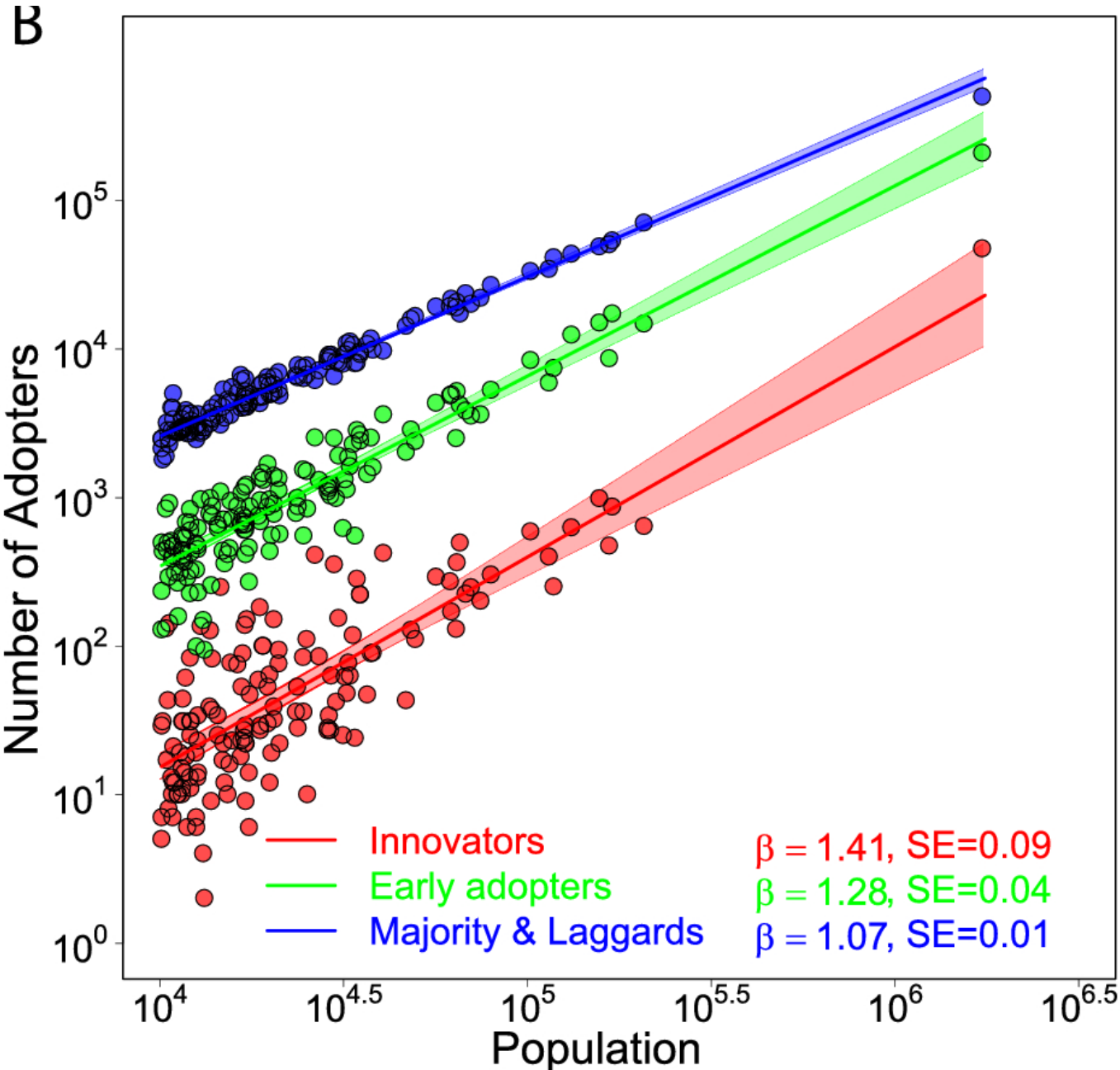


The role of distance

Distance only slightly decreases the probability of invitation in early phases

Diffusion becomes local in later phases.

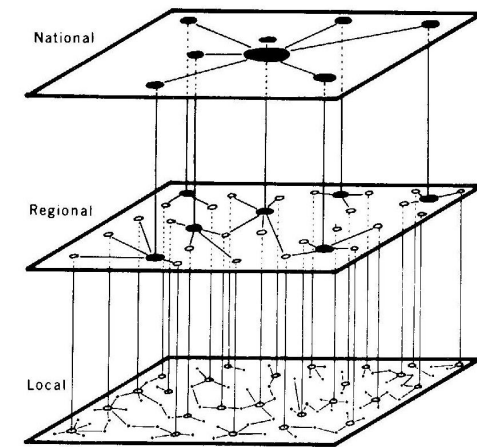




The role of size

Innovators and Early Adopters concentrate in cities.

Majority and Laggards are proportional to town size.



Adoption Rate

(Bass Curve)

$$\frac{dy_a(t)}{dt} = (p_a + q_a y_a(t))(1 - y_a(t))$$

$$y_a(t) = m \frac{1 - e^{-(p_a + q_a)t}}{1 + \frac{q_a}{p_a} e^{-(p_a + q_a)t}}$$

of adopters

Innovation Par.

$y_a(t)$

p_a

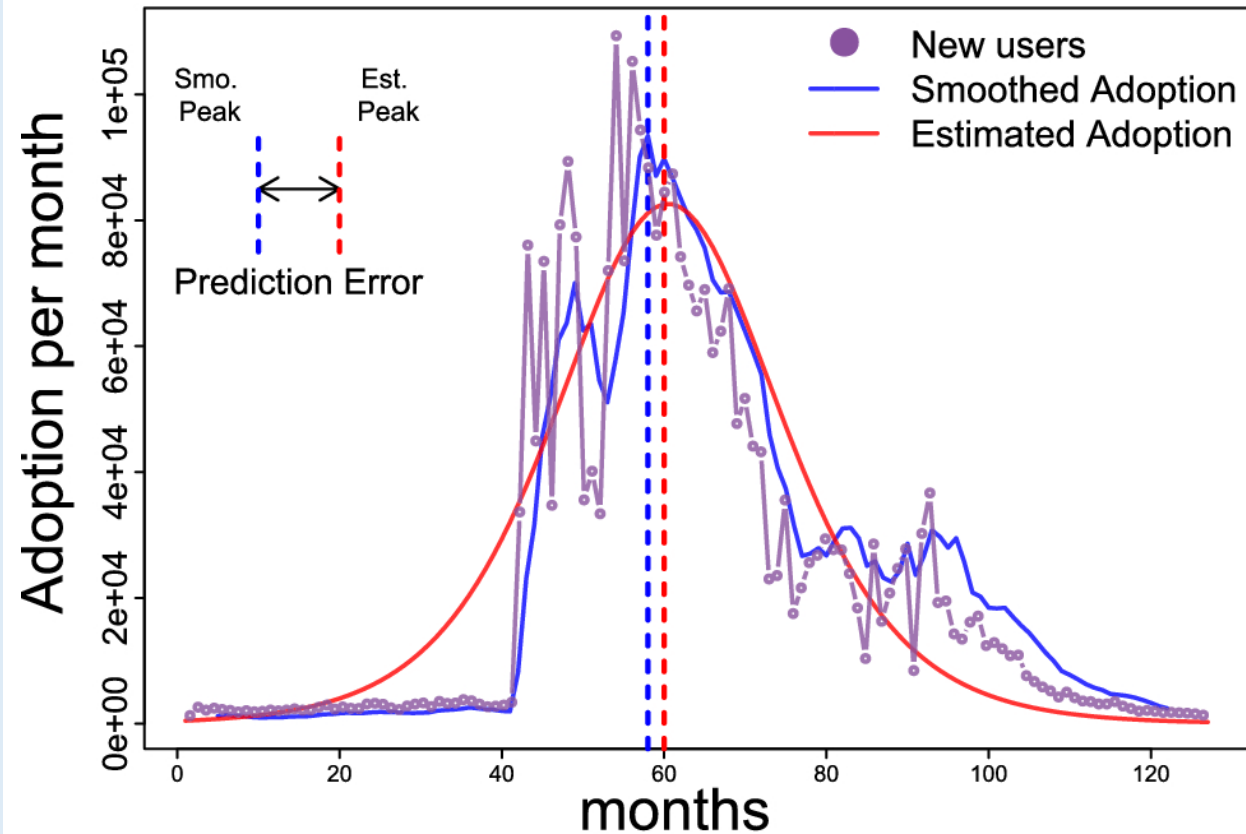
t

q_a

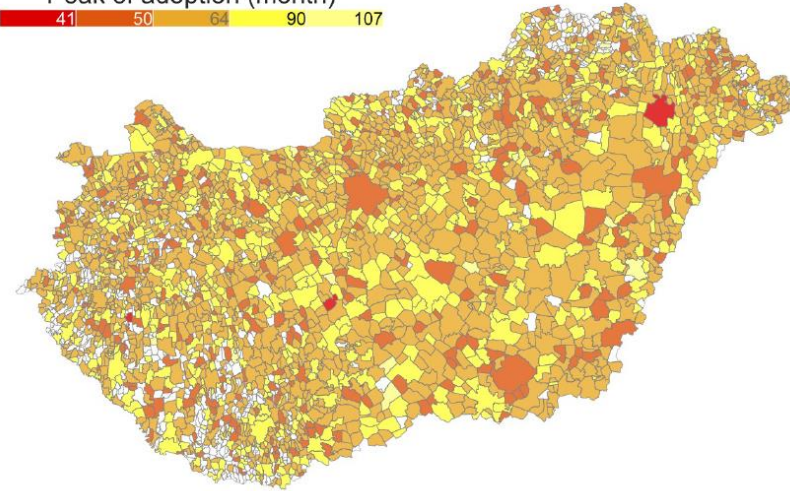
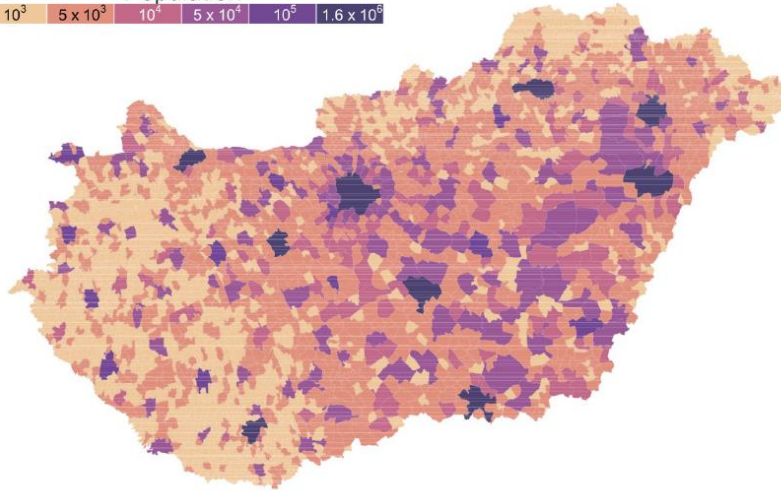
Time (months)

Imitation Par.

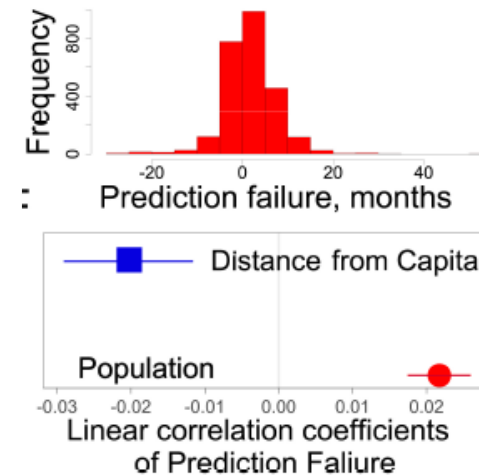
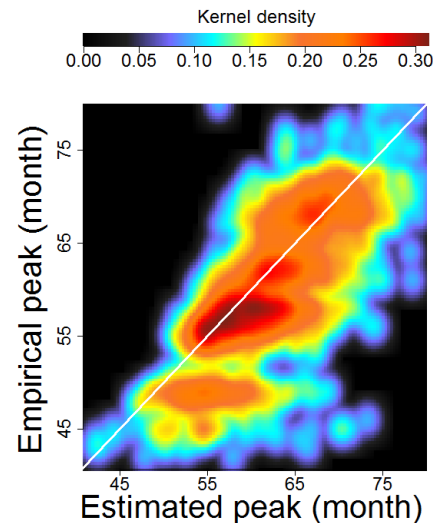
A



Local adoption peaks and prediction errors

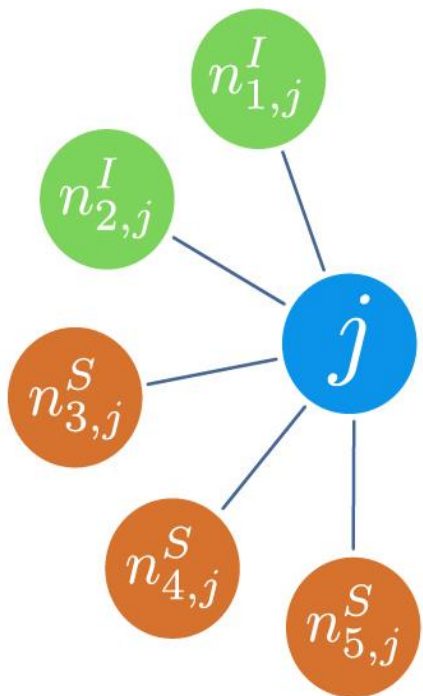


$$T_{peak-t}^i = \frac{\ln p_a^i + \ln q_a^i}{p_a^i + q_a^i}$$

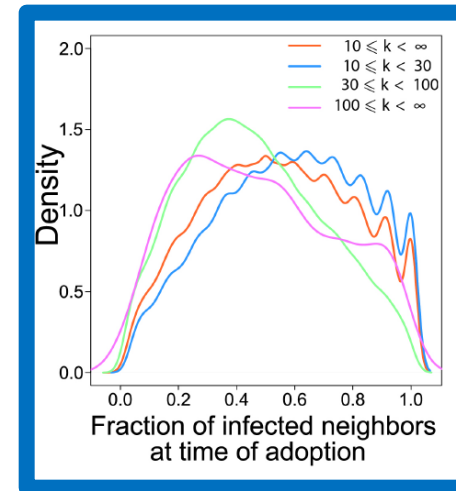


Agent-based model of diffusion in the network

- We fix the network.
- Inactives are not removed.
- 10% sample (300 K users)
- Sampling stratified by towns and network moduls



Innovation or
Marketing
parameter



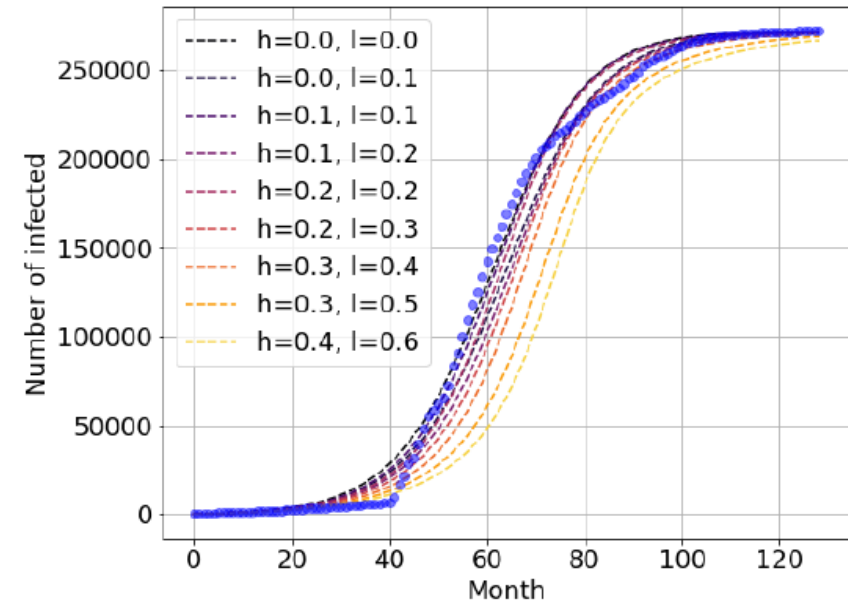
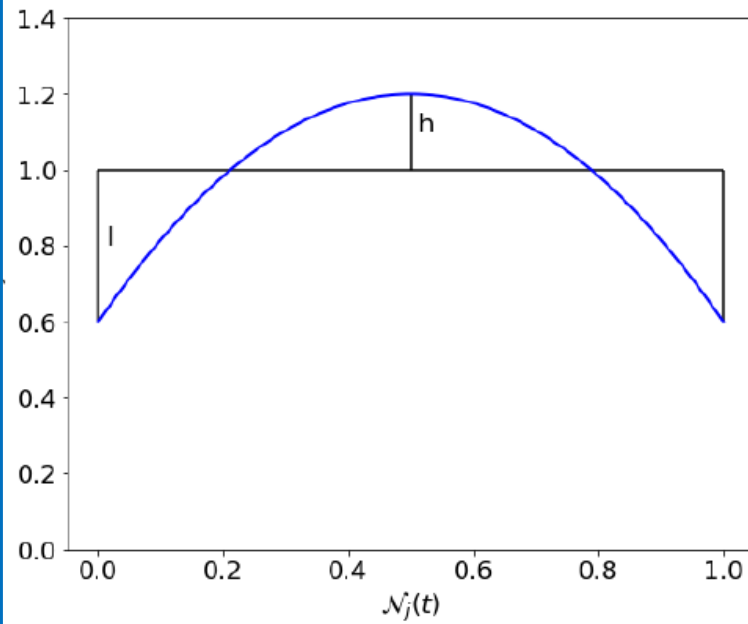
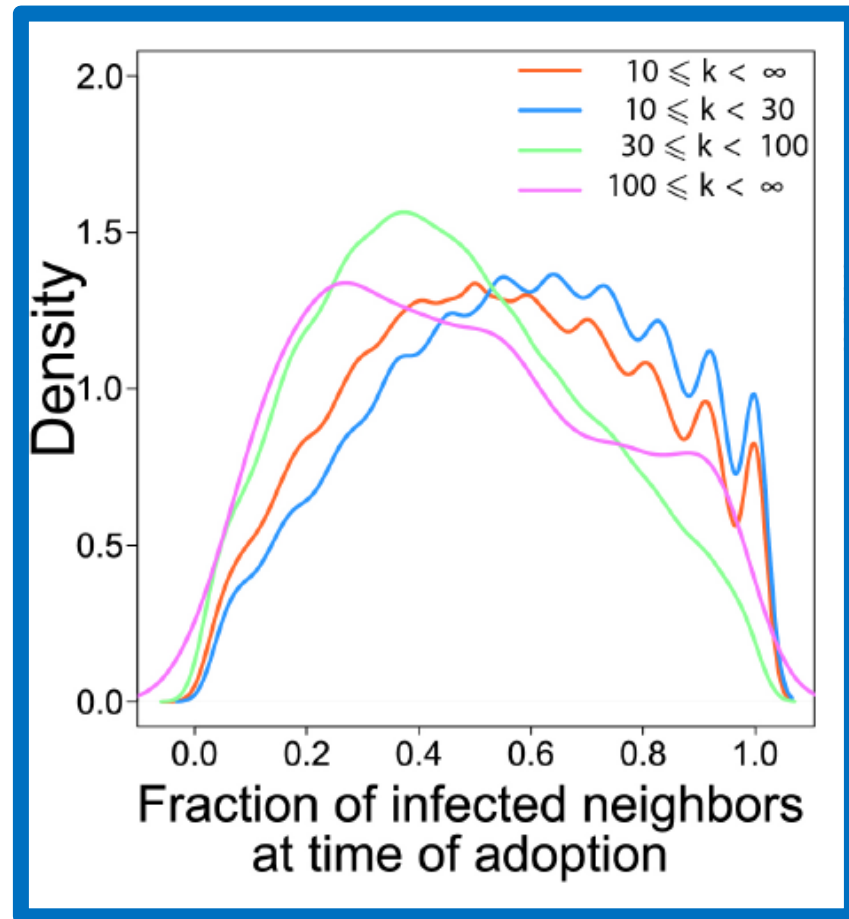
Network effect

$$\mathcal{N}_j(t) = \frac{\#n_j^I(t)}{\#n_j^I(t) + \#n_j^S(t)}$$

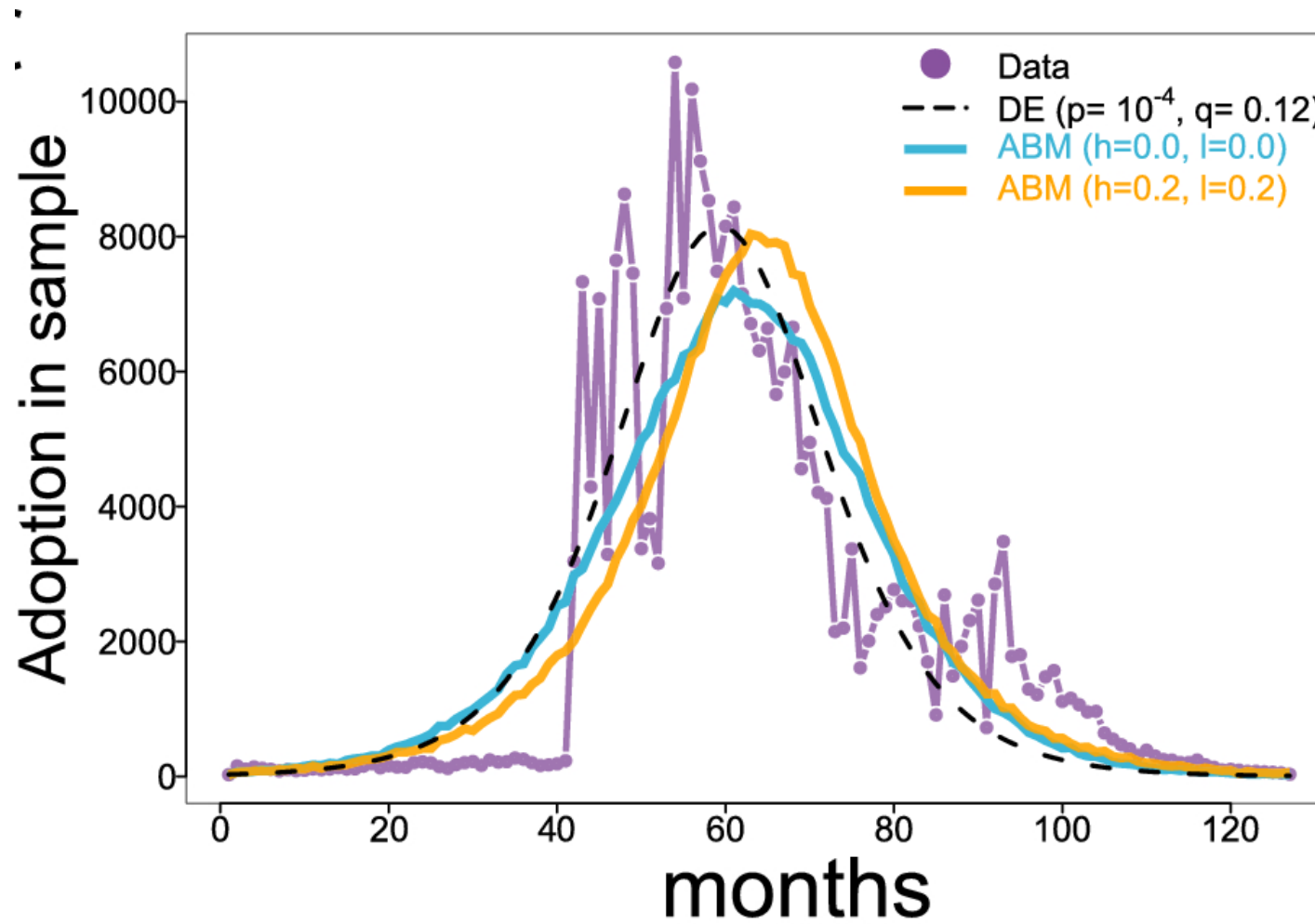
$$F_j(t+1) = \begin{cases} I & \text{if } U(0,1)_{jt} < \hat{p}^{\text{ABM}} + T(\mathcal{N}_j(t), h, l) \times \mathcal{N}_j(t) \times \hat{q}^{\text{ABM}} \\ S & \text{otherwise} \end{cases}$$

Imitation parameter

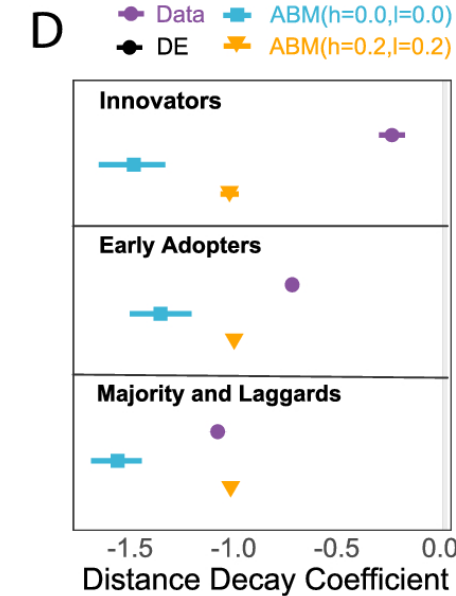
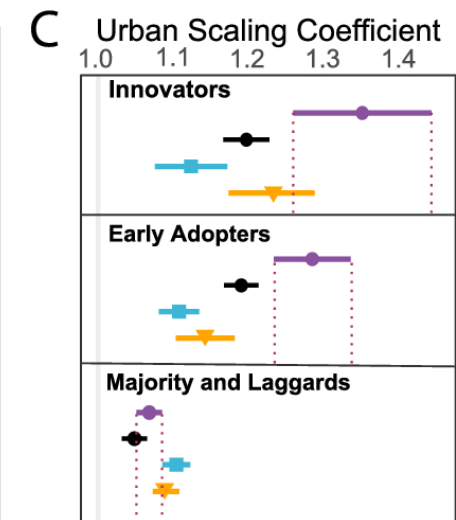
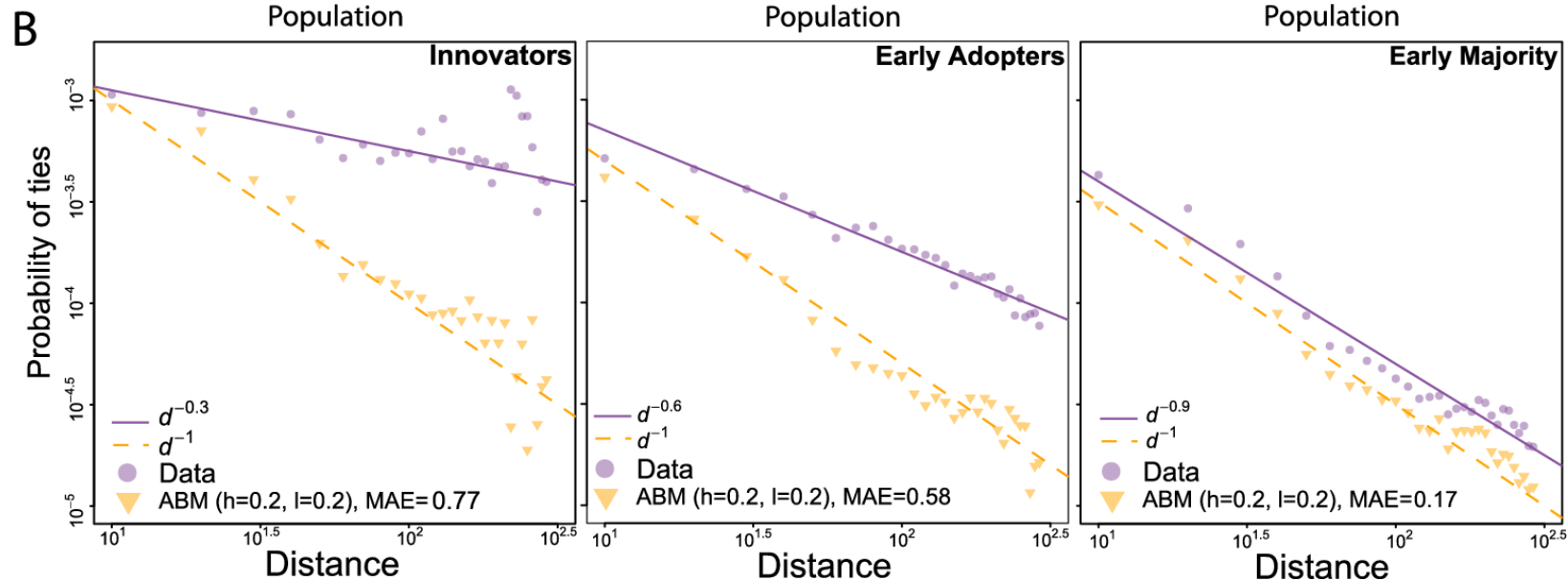
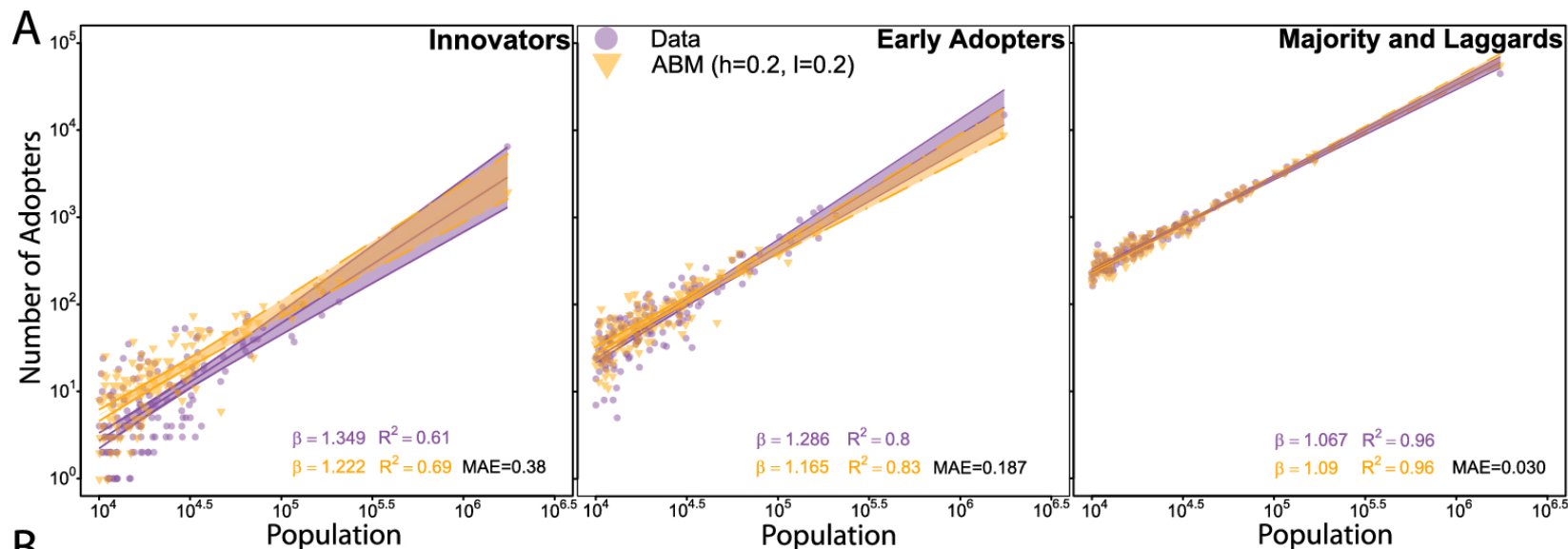
Threshold parameters



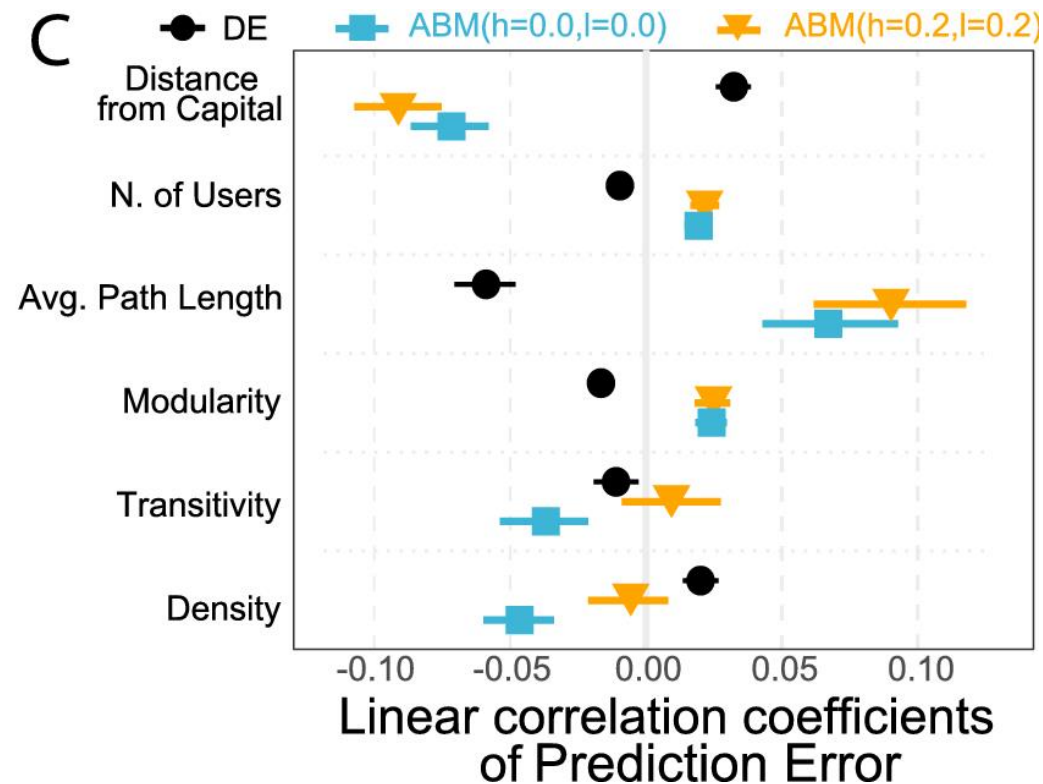
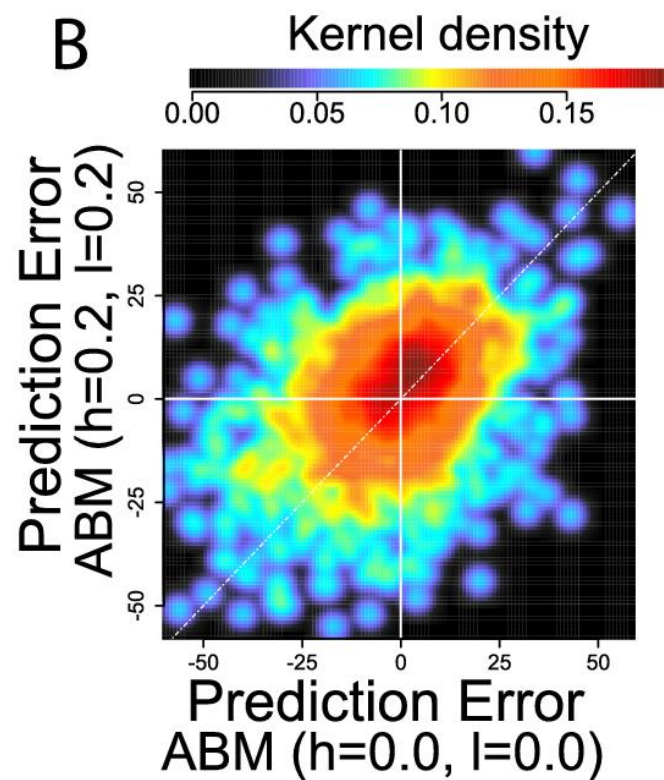
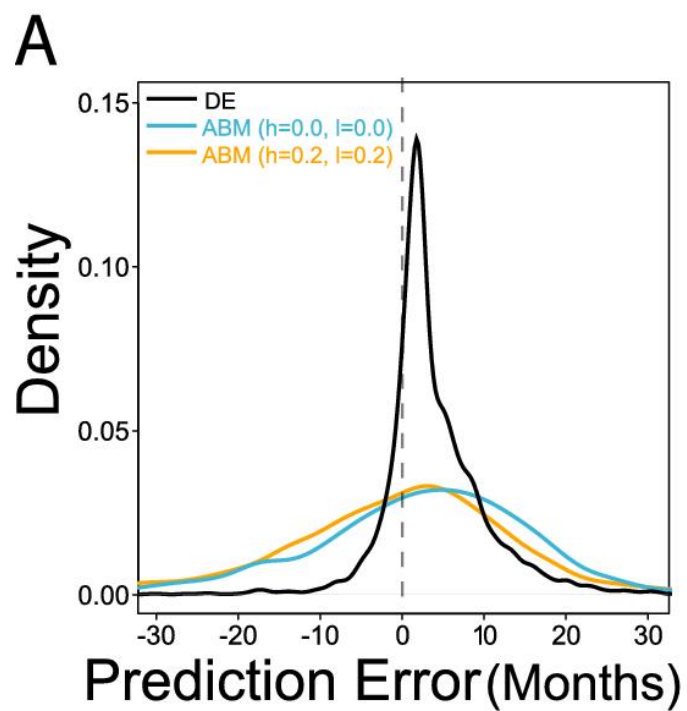
Results



Results



Results





CHURN

